Geophysical Journal International

Geophys. J. Int. (2012) **189,** 204–228 doi: 10.1111/j.1365-246X.2012.05357.x

Palaeointensity and palaeomagnetic study of Cretaceous and Palaeocene rocks from Western Antarctica

V.V. Shcherbakova,[1] V.G. Bakhmutov,[2] V.P. Shcherbakov,[1] G.V. Zhidkov[1] and V.V. Shpyra[2]

[1] *Geophysical Observatory "Borok" of Physics of the Earth of RAS, Russia. E-mail: valia@borok.yar.ru*
[2] *Institute of Geophysics of the National Academy of the Sciences of Ukraine, Ukraine*

Accepted 2011 December 13. Received 2011 December 13; in original form 2011 August 7

SUMMARY

A combined palaeodirectional and palaeointensity study of a representative collection of plutonic rocks from the Antarctic Peninsula batholith from the western part of the Antarctic Peninsula, near the Ukrainian Antarctic base 'Academik Vernadsky' were carried out. Petrographically, the collection includes gabbros, diorites and quartz diorites, tonalities, granodiorites and granites. The ages of igneous complex emplacement vary from 50 to 117 Ma with most of the rocks belonging to the Cretaceous Normal Superchron. The characteristic remanent magnetizations were isolated by stepwise thermal demagnetization over the temperature interval 440–590°C and their intensities amount to 95 per cent of the NRM. The geographic positions of palaeopoles do not contradict the 'key poles' of the Antarctic Peninsula between 90 and 60 Ma. A significant part of the collection was subjected to Coe-modified Thellier palaeointensity experiments with the pTRM checks, which yielded seven reliable palaeointensity determinations for seven different locations. The obtained VDMs are relatively low for all sites, being on average about half of the present day VDM. The analysis of available palaeointensity data for the Cretaceous, Miocene and Middle Jurassic indicates the existence of strong correlations between the mean VDM and VDM scatter versus the rate of reversals. However, due to the shortage of data, the correlations are not significant at the 5 per cent significance level.

Key words: Palaeointensity; Palaeomagnetism applied to tectonics; Reversals: process, timescale, magnetostratigraphy; Rock and mineral magnetism.

1 INTRODUCTION

Variations of the geomagnetic field intensity over geological time are linked to processes of geodynamo generation taking place in the Earth's deep interior. Since the work by Cox (1968), an ongoing goal in palaeomagnetism is the search for a possible relationship between the frequency of reversals and palaeointensity. As it was realized long ago, palaeointensity study of the Cretaceous period with its close to zero reversal rate inside the Cretaceous Normal Superchron (CNS) can provide key evidence to solve this problem.

The first extensive Thellier palaeointensity determinations for the Cretaceous were reported by Bol'shakov & Solodovnikov (1981) who claimed that palaeointensity at that period was 2–3 times less than the present geomagnetic field intensity. Prévot *et al.* (1990) extended this trend over the whole Mesozoic era. The hypothesis of the Mesozoic Dipole Low was supported by Perrin & Shcherbakov (1997) on the basis of statistical analysis of the palaeointensity determinations published to that time. Later, based on palaeointensity determinations from submarine basaltic glasses, Juarez *et al.* (1998) concluded that the average dipole moment of the Earth over the past 160 Ma was only half as great as the present day field, so that the Mesozoic field was not actually 'low', but of average intensity. On the other hand, high geomagnetic intensities for the CNS have been reported (Tarduno *et al.* 2001, 2002, 2006; Goguitchaichvili *et al.* 2002b; Tauxe & Staudigel 2004; Tarduno & Cottrell 2005; Tauxe 2006). When considered results of collectively, recent studies by Shcherbakova *et al.* (2007, 2008, 2009, 2011) still favour the hypothesis of low field in the Cretaceous.

Here we report new palaeomagnetic and palaeointensity data obtained from the Cretaceous and Palaeocene rocks collected in Western Antarctica. This region is of particular interest because during the Cretaceous this continent was located at high latitudes and the great majority of the Cretaceous palaeointensity data are obtained from low or middle palaeolatitude locations. This study that concentrates on the investigation of high palaeolatitude region, can provide a more robust data set for the Cretaceous.

© 2012 The Authors
Geophysical Journal International © 2012 RAS

2 DETERMINATION OF PALAEODIRECTIONS

2.1 Geology and experimental procedure

In contrast to East Antarctica, which mainly consists of Precambrian rocks, West Antarctica is a geological entity with Phanerozoic aged complexes (Grikurov 1973; Dalziel & Elliot 1982). The Antarctic Peninsula (AP) is the largest tectonic block of West Antarctica and the most accessible area to investigate the tectonic processes that occurred along the southeast Pacific continental margin. Recent geophysical and geological studies suggest that the AP is a composite magmatic arc comprising two or three separate terrains that were accreted and sutured along the Gondwana margin in the mid-Cretaceous (Vaughan & Storey 2000; Ferraccioli *et al.* 2006).

Approximately 80 per cent of rocks exposed on the islands of the AP shelf as well as on the mainland are formed by plutonic rocks of the AP batholiths (Leat *et al.* 1995) and volcanic rocks of the AP Volcanic Group (Thomson & Pankhurst 1983). The plutonic rocks forming the AP batholiths were emplaced within the period ∼240 to 10 Ma with an Early Cretaceous peak of activity (Leat *et al.* 1995). The Early Cretaceous plutonic rocks of gabbro-granite composition (with the prevalence of diorites) constitute the widespread plutons and batholiths (Leat *et al.* 1995; Willan & Kelley 1999). The acidic rocks are younger than the basic ones (Rex 1976). The volcanic rocks of the AP Volcanic Group occur as veins, dykes and flows represented by andesites, diabases, basalts, rhyolites, dolerites and dacites (Weaver *et al.* 1982; Riley *et al.* 2001).

Most palaeogeographic reconstructions made for the AP and South America are based on seafloor magnetic anomalies. However, palaeomagnetic studies have also been carried out to provide constraints on the relative positions of South America, the AP and East Antarctica. Palaeomagnetic results for the AP have already been reported in a number of studies (Blundell 1962; Dalziel *et al.* 1973; Kellogg & Reynolds 1978; Valencio *et al.* 1979; Kellogg 1980; Longshaw & Griffiths 1983; Watts *et al.* 1984; Grunow 1993; Pares & Dinares 1999; Nawrocki *et al.* 2010; Poblete *et al.* 2011). Some of these studies (Blundell 1962; Grunow 1993; Poblete *et al.* 2011) provide detailed palaeomagnetic and geological information for this region that facilitates the combination of the data presented here with the previously published results.

A representative collection of rocks was sampled in the area close to the Ukrainian Antarctic station 'Akademik Vernadsky' (64.27°W, 65.25°S) whose location is marked by 'V' in Fig. 1(b). About 400 oriented samples from 24 sites were drilled or collected as hand samples, both on the Argentine Islands archipelago and on the adjoining western coast. Sampling at each site covered the maximum accessible (free from ice) area. All samples were collected from blocky, solid and relatively unaltered intrusions and comagmatic dykes thus maximizing the suitability of rocks for palaeomagnetic experiments. The natural remanent magnetization (NRM) of samples is typically less than 1 A m^{-1}, which is too small to affect the orientation with a magnetic compass. A correction to the field measurements has been made to offset the local magnetic declination 16°E. Later we mainly discuss the results concerning the sites where reliable palaeointensity determinations are obtained. The locations of these sites are shown in Figs 1(b) and (c).

The ascribed radiometric ages are based on previously published data (Rex 1976; Fleming & Thomson 1979; Pankhurst 1982, 1983; Leat *et al.* 1995; Tangeman *et al.* 1996; Guenthner *et al.* 2010) as well as on the recent U–Pb zircon results for rocks studied in this work (Gladkochub *et al.* 2009).

The experiments for determining of palaeodirections were carried out in the palaeomagnetic laboratory of the Institute of Geophysics of the National Academy of Science of Ukraine in Kiev. The collection was stored in a magnetically shielded room and all measurements were made there. The samples were cut to standard cylinders or to cubes 20 mm in edge length. The vectors of the characteristic remanent magnetization (ChRM) have been isolated by both stepwise thermal and alternating field (AF) demagnetization. Stepwise thermal demagnetization up to 600–610°C in 15–20 steps was carried out using a MMTD80 oven. After each heating step the susceptibility at room temperature T_r was measured to monitor possible mineralogical changes. A JR-6 spin magnetometer and a MFK1 kappabridge were used for magnetization and susceptibility measurements, respectively. Some duplicate specimens were subjected to AF demagnetization using a LDA-3A demagnetizer and these results confirmed the thermal demagnetization data. Petrography observations of thin sections were carried out to classify the geological type of rocks.

Results of magnetic measurements were processed with the Remasoft 3.0 program (Chadima & Hrouda 2006). The ChRM vector of each sample was determined using the least-squares method (Kirschvink 1980). Demagnetization data were plotted on vector endpoint diagrams and ChRM directions were calculated from linear segments converging toward the origin. The palaeodirections obtained were considered as meaningful if they have been defined by three or more collinear demagnetization steps with a maximum angular deviation less than 5°.

2.2 Sampling, radiometric dating and determination of palaeodirections

2.2.1 Western coast of Antarctic Peninsula

Cape Tuxen and Deliverance point. Banded hornblende gabbros crop out on the N–W point of Cape Tuxen (Curtis 1966). Granodiorite dykes, which intrude the gabbros are observed in cliff faces. U–Pb dating of quartz diorite from prominent point of Cape Tuxen yields an age of 85.2 ± 0.7 Ma (Tangeman *et al.* 1996; Table 1). U–Pb dating of quartz diorite from the base of Mount De Maria and from granodiorite collected approximately 1 km southeast of Cape Tuxen yield ages of 84.8 ± 0.5 Ma and 84.5 ± 0.9 Ma, respectively (Tangeman *et al.* 1996). Recent U–Pb dating of zircon crystallization in the gabbros from Cape Tuxen yielded an age of 88.1 ± 1.1 (Gladkochub *et al.* 2009).

Palaeomagnetic studies of these rocks have been performed previously by Blundell (1962) and Grunow (1993). Palaeodirections of the NRM were determined as $D = 9°$; $I = –77°$ [mean of eight samples from Cape Tuxen area (Blundell 1962)] and as $D = 4.7°$; $I = –75.2°$ [mean of three sites of granodiorite and three sites of gabbros (Grunow 1993)].

Along the coastline of the AP we collected gabbros on the N–W prominent point of Cape Tuxen (sampling area ≈ 100 × 50 m), granodiorites on the south coast of Waddington Bay (3 km to southeast from Cape Tuxen, sampling area 30 × 70 m) and granodiorites of Deliverance point (3 km to south along the sea coast, sampling area 50 × 100 m). Our palaeomagnetic study of these rocks revealed mainly a single N-polarity component of NRM (Fig. 2a). The Cape Tuxen site direction ($D = 2.8°$; $I = –79.2°$) is in agreement with the data reported by Blundell (1962) and Grunow (1993). The two others sites have similar inclinations, but a small scattering of declinations (Table 2).

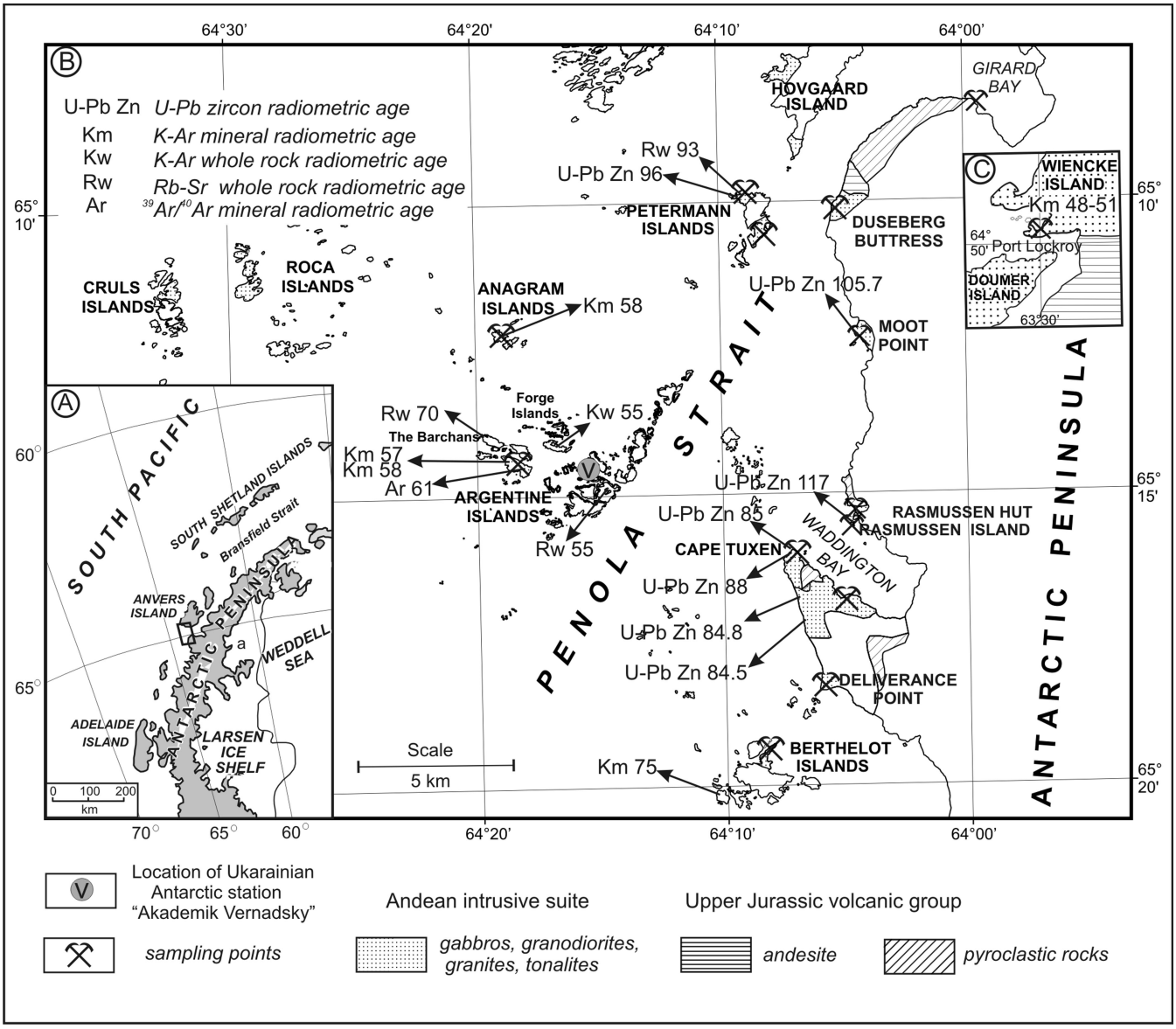


Figure 1. (A) Map of the Antarctic Peninsula. The small rhomb in the centre depicts the location of the sampling site, details of the site's locations are shown in (B) and (C). (B) Argentine Islands archipelago area and the Western Coast of Antarctic Peninsula. (C) Wiencke Island, near the Port Lockroy. Geological setting modified from Curtis (1966). Radiometric ages (Ma) are given in Table 1.

Moot point. A promontory of the AP along Penola Strait and a group of adjacent small islands are known as Moot Point. The entire area is composed of medium grained, biotite + hornblende bearing diorite, which are cut by nearly vertical north–south trending mafic dykes.

U–Pb dating of quartz diorite yields an age of 105.7 ± 0.7 Ma (Tangeman *et al.* 1996). There are no age determinations from the mafic dykes, but the dykes yield palaeomagnetic directions indistinguishable from the ones from the diorite (Grunow 1993) and they do not cut the Cape Tuxen rocks. These lines of evidence imply that the dykes are older than 85 Ma and possibly comagmatic with the diorite.

Previous investigations by Grunow (1993) suggest two distinct components of NRM from the diorites. The A-component mean direction, obtained from five sites, had $D = 23.8^\circ$, $I = -71.2^\circ$. In some diorite samples a second, B-component, was detected (13 samples, $D = 297.2^\circ$, $I = -76.3^\circ$).

Our sampling was made on a small area (20 × 10 m) of sea coast of the promontory of the AP (diorite) and in the western part of the biggest unnamed island located several 100 m opposite to the promontory (granodiorite). Only N-polarity components of NRM in diorite and granodiorite samples ($D = 8.8^\circ$, $I = -65.3^\circ$ and $D = 39^\circ$, $I = -63.8^\circ$, respectively) have been revealed (Fig. 2b, Table 2). These directions are close to the A-component found by Grunow (1993), but have shallower inclinations. The directions of NRM from granodiorite samples are characterized by scattering $\alpha_{95}^\circ = 14.3$ and display about 30° difference in declination relative to diorites from mainland.

Rasmussen Hut and Rasmussen Island. The exposures on the promontory of headland near Rasmussen Hut and on Rasmussen Island, located 100 m from the AP, are composed of different diabase, granodiorite and granite cut by comagmatic mafic dykes. The coarse grained pink granite on Rasmussen Island yields a U–Pb

Table 1. Summary of age determinations of the AP batholith plutonic rocks as reported in the original references.

Locality	Lat., Long	Rock types	Radiometric method	Mineral analysed	Analyses reference number	Age (Ma)	Reference
Cape Tuxen and Deliverance Point	64°07′W, 65°16′S	Quartz diorite	U–Pb	Zircon	AP90–11H	85.2 ± 0.7	Tangeman *et al.* (1996)
	64°071′W, 65°161′S	Gabbro	U–Pb	Zircon		88.1 ± 1.1	Gladkochub *et al.* (2009)
	64°066′W, 65°171′S	Quartz diorite	U–Pb	Zircon	AP90–11J	84.8 ± 0.5	Tangeman *et al.* (1996)
	64°06′W, 65°18′S	Granodiorite	U–Pb	Zircon	AP90–11K	84.5 ± 0.9	Tangeman *et al.* (1996)
Moot Point	64°045′W, 65°123′S	Quartz diorite	U–Pb	Zircon	AP90–11A	105.7 ± 0.7	Tangeman *et al.* (1996)
Rasmussen Island	64°045′W, 65°155′S	Granite	U–Pb	Zircon	AP90–11F	117.0 ± 0.8	Tangeman *et al.* (1996)
		Pink granite	Rb–Sr			128.0 ± 3.0	Pankhurst (1982)
Petermann Island	64°09′W, 65°10′S	Diorite/ granite	Rb–Sr whole rock			93.0 ± 8.0	Pankhurst (1982)
		Quartz diorite	U–Pb	Zircon		95.9 ± 1.0 96.1 ± 0.7	Gladkochub (2010)
Berthelot Islands	64°10′W, 65°19′S	Diorite	K–Ar	Pyroxene	IDB597	73.0 ± 6.0	Rex (1976)
Argentine Islands	64°15′W, 65°15′S	Diorite/ granodiorite/ aplite	Rb–Sr			55.0 ± 3.0	Pankhurst (1982)
Forge Islands	64°18′W, 65°14′S	Hornblendite	K–Ar	Hornblende	IDB801	54.0 ± 2.0	Rex (1976)
Anagram Islands	64.192′W, 65.123′S		K–Ar			58.0	Fleming & Thomson (1979)
The Barchans, South Island	64°20′W, 65°14′S	Quartz diorite	K–Ar	Biotite	IDB574 IDB701 IDB583	56 ± 2; 56 ± 2; 57 ± 2	Rex (1976)
The Barchans, South Island	64°18′W, 65°145′S	Granodiorite	$^{39}Ar/^{40}Ar$	Biotite		60.9 ± 0.1	Gladkochub *et al.* (2009)
The Barchans, Western Island		Granodiorite	Rb–Sr whole rock			70	Fleming & Thomson (1979)
Goudier Island, Port Lockroy	63°31′W, 64°50′S	Quartz diorite	K–Ar	Biotite hornblende	IDB514 IDB539	51 ± 2; 51 ± 2	Rex (1976)
Port Lockroy	63°30′W, 64°49′S	Quartz diorite	K–Ar	Biotite hornblende biotite	IDB687 IDB711 IDB603	48 ± 2; 51 ± 2; 49 ± 2	Rex (1976)

zircon age of 117.0 ± 0.8 Ma (Tangeman *et al.* 1996). Stepwise thermal demagnetization performed on these rocks by Grunow (1993), indicated a single component NRM for both the granite and mafic dykes from Rasmussen Island with mean direction of $D = 14.4^\circ$, $I = -72.1^\circ$ from five sites.

For the purpose of this work granodiorite and diabase dykes from the headland near Rasmussen Hut (sampling area 100 × 100 m) and two subparallel mafic dykes on the Rasmussen Island were sampled. Our results confirm the presence of the univectorial N-polarity component in islet dykes ($D = 339.8^\circ$, $I = -82.3^\circ$; Fig. 2c) and headland granodiorites + dykes ($D = 21.9^\circ$, $I = -76.1^\circ$). The mean direction of all collected samples in this area ($D = 12.4^\circ$, $I = -78.9^\circ$) agrees with the determination by Grunow (1993).

Duseberg Buttress and Girard Bay. The most spectacular exposures of volcanic rocks of the AP Volcanic Group are located on the mainland cost of Penola Strait and on the shores bordering Lemaire Channel where volcanic rocks crop out between Duseberg Buttress and Girard Bay over a distance of about 1.5 km. The volcanic rocks had been folded into an asymmetrical syncline with the axis pointing N–W to S–E and they have been intruded by diorite about 0.8 km north of Duseberg Buttress. The sampling in this region was carried out on two points opposite of Petermann Island (diorites of Duseberg Buttress) and SW of Girard Bay (gabbro).

No age and palaeodirectional determinations are known for these sites, but the analysis of geological structures indicates similarity between diorites from Duseberg Buttress and diorites from the Moot

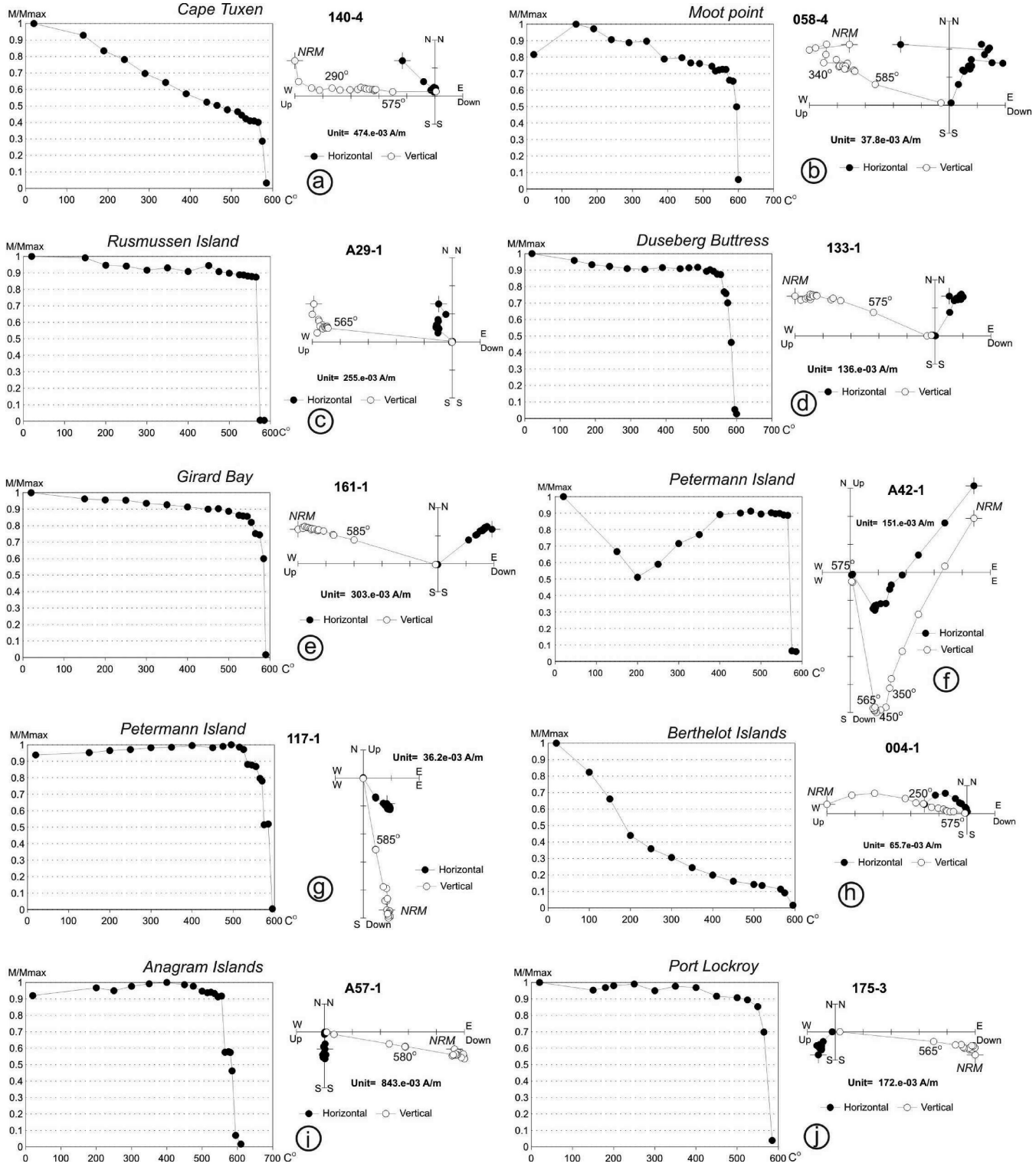


Figure 2. Examples of stepwise thermal demagnetization curves of samples from the Western Coast of Antarctic Peninsula (a–e) and adjacent islands (f–j). For each sample the normalized NRM intensity decay curve (left-hand side) and the orthogonal plot (right-hand side) are shown.

Point, as well as between gabbros on Girard Bay and gabbros on Petermann Island and Cape Tuxen. Based on this similarity we assume that the ages of the rocks at these sites correspond to the CNS.

A single N-polarity component of NRM was detected in Duseberg Buttress diorites ($D = 31.5^\circ$, $I = -67.2^\circ$; Table 2, Fig. 2d). The gabbros from the SW part of Girard Bay also have a single N-polarity component with $D = 53.4^\circ$, $I = -65.1^\circ$ (Fig. 2e). This declination is higher than in other sites (Table 2) probably due to local tectonics. The bedding correction here is unclear as we do not know whether the gabbros intruded after or before the deformation of the area.

Table 2. Palaeomagnetic directions and virtual geomagnetic poles.

	Site	Samples	Lithology	NRM (A m^{-1})	Component	n/N_t	D°	I°	K	α_{95}°	VGP Lat.°S	VGP Long.°E	Component polarity
Cape Tuxen and Deliverance Point													
1	Cape Tuxen (64.11°W, 65.27°S)	140–153; A23-A27	Gabbro	0.8–4.7/2.5		15/19	2.8	–79.1	128	3.4	86.1	280.6	*N*
2	Deliverance Point (64.08°W, 65.31°S)	84–94	Granodiorite	0.3–0.5/0.4		13/13	28.5	–77.5	161	3.3	78.3	222.5	*N*
3	Waddington Bay (64.05°W, 65.29°S)	96–101; 105–106	Granodiorite	0.1–0.7/0.4		8/8	13.2	–76.4	307	3.2	84.3	201.5	*N*
Moot Point (64.07°W, 65.2°S)													
4	Moot Point (mainland)	53–60	Diorite	0.07–0.5/0.2		8/8	8.8	–65.3	189	4.0	71.6	135.0	*N*
5	Moot Point (island)	41–45; 50–52	Granodiorite	0.05–0.6/0.3		5/8	39.0	–63.8	30	14.3	61.1	182	*N*
Rasmussen Hut and Rasmussen Island (64.07°W, 65.25°S)													
6	Rasmussen Island	A28–A37	Diabase	0.08–2.8/1.06		10/10	339.8	–82.3	151	3.9	78.3	322.2	*N*
7	Rasmussen Hut	021–035; 154–157	Granodiorite Diabase (dyke)	0.3–1.0/0.6; 2.6–11.2/5.4		18/20	21.9	–76.1	283	2.1	80.4	206.3	*N*
Duseberg Buttress and Girard Bay													
8	Duseberg Buttress (64.1°W, 65.16°S)	131–137	Diorite	0.5–0.9/0.7		6/7	31.5	–67.2	336.4	3.7	67.7	178.2	*N*
9	Girard Bay (64.0°W, 65.13°S)	158–165	Gabbro	0.3–2.0/1.1		8/8	53.4	–65.1	496	2.5	56.7	199.8	*N*
Petermann Island (64.15°W, 65.17°S)													
10	Petermann Island	107–120; A38-A45; 941; 942	Gabbro	0.2–6.0/1.7	Comp B	24/24	151.0	76.0	168	2.3	77.4	20.4	*R*
10’		107—120; A38-A45; 941; 942	Gabbro		Comp A	19/24	6.7	–67.9	22.8	7.2	75.3	132.8	*N*
Berthelot Islands (64.13°W, 65.33°S)													
11	Berthelot Islands	16–20; A58–A64	Gabbro	0.2–1.0/0.5		12/12	359.9	−76.2	184.7	3.2	88.5	114.3	*N*
12		001–015	Granodiorite	0.06–0.3/0.16		13/14	9.2	−78.4	55	5.6	85.7	242.4	*N*
Archipelago Argentine Islands. The Barchans and Anagram Islands													
13	Anagram Islands (Nob Island) (64.32°W, 65.2°S)	A51–A57	Gabbro	2.6–12.0/3.7		7/7	176.8	77.5	541	2.6	88.4	351.8	*R*
14	The Barchans (64.32°W, 65.23°S)	38–40; 835–856	Granodiorite	0.04–0.4/0.17	Comp B	22/24	151.1	81.8	102	3.1	76.9	332.1	*R*
14′					Comp A	12/24	334.7	−81.4	34	7.6	78.2	332.7	*N*
Port Lockroy (63.5°W, 64.83°S), Wiencke Island													
15	Port Lockroy	167–180	Granodiorite	0.1– 8.0/1.7		9/12	219.5	84.3	338	2.8	72.1	272.6	*R*

Note: n/N$_t$ is the numbers of successful samples used for site mean calculation/total number of studied samples; NRM column shows the range of NRM values (above the line), the value below the line is the mean NRM; D° and I° are site-mean declination and inclination of the isolated ChRM component, respectively; α_{95}° is the 95% confidence cone around the mean direction; K is the Fisherian concentration parameter of the individual ChRM vectors; VGP is virtual geomagnetic pole; dp and dm are the semi-axes of the 95% confidence ellipse of the pole; N is the normal, R is the reversed polarity. A declination correction of 16° has been applied to all results.

2.2.2 Adjacent islands

Petermann Island. Gabbros and quartz-diorites crop out in the N–NW part of the island, and the sampling area stretches about 800 m along the coastline. Results of Rb–Sr whole rock radiometric study reported by Pankhurst (1982) yield an age of 93 Ma while U–Pb dating of zircon from quartz-diorite gave an age of 96 Ma (Gladkochub 2010, personal communication).

In some samples two distinct components (A and B) of NRM were present. The low temperature normal polarity A component with $D = 6.7^\circ, I = -67.9^\circ$ corresponds to the blocking temperatures 150–400 °C, but in the high temperature interval a reversed polarity component B with the mean direction $D = 151.0^\circ$, $I = 76.0$ is observed (Fig. 2f). We note that the reversed component B is clearly distinguished for all investigated samples as shown in Table 2 and illustrated in Fig. 2(g).

Berthelot Islands. The northern part of the Berthelot Islands is comprised of gabbros and diorites. The K–Ar data obtained for diorite from the southern part of the islands yield the age 73 ± 6 Ma (Rex 1976).

The sampling area covers 50 × 300 m on the NE promontory, which was free of ice. Single N polarity components were isolated in both gabbros and granodiorites (Fig. 2h) with $D = 359.9^\circ$, $I = -76.2^\circ$ and $D = 9.2^\circ, I = -78.4^\circ$, respectively (Table 2), which are close to the mean palaeodirection ($D = 358^\circ$, $I = -72^\circ$) obtained from eight gabbro samples by Blundell (1962).

Argentine Islands archipelago. A number of islands located on the area 6 × 6 km have a common geographic name the Argentine Islands. The two biggest (Galindez and Winter) islands are well known as the Ukrainian Antarctic station 'Akademik Vernadsky' (the former British base 'Faraday') and an old British base, respectively, are located there. The archipelago is separated from the mainland by the deep Penola Strait, which geologically is a fault between the AP and Western Islands. The western part of the Argentine Islands archipelago and other western archipelagos are composed by plutonic rocks, which outcrop on the Barchans Islands, Forge Islands, Anagram Islands and in the area further to the west. The eastern part of the archipelago is composed of AP Volcanic Group rocks (mainly andesite lavas and associated pyroclastic rocks, which have been metamorphosed and metasomatized by the Andean intrusive suite) and separated from western plutonic rocks by sublongitudinal contact zone about 1000 m in width.

The object of our research on the Argentine Islands archipelago was only plutonic rocks. Gabbros, norites and tonalites occur mainly in the Anagram Islands and granodiorites crop out principally in the Barchans and the Forge Islands. A series of K–Ar and Rb–Sr data indicate that the age of these rocks is Palaeocene (55–70 Ma, Table 1). Dating of the quartz-diorite from the south island of Barchans Islands by K–Ar mineral radiometric ages (biotite) yields 56–57 Ma (Rex 1976). The K–Ar age of the hornblendite from Forge Islands is 54 ± 2 Ma (Rex 1976) and the Rb–Sr dating for the North Barchans Island rocks yielded an age of 70 Ma (Fleming & Thomson 1979). The K–Ar age from Anagram Islands is 58 Ma (Fleming & Thomson 1979). Recent $^{39}Ar/^{40}Ar$ dating of biotite from quartz-diorite of the South Barchans Island yields the age of 60.9 ± 0.1 Ma (Gladkochub *et al.* 2009).

The samples were taken on the three biggest Barchans Islands (granodiorites within an area of 1.5 × 1.0 km) and on the biggest Nob Island of Anagram Islands archipelago (gabbros, sampling area 300 × 300 m). Palaeomagnetic data from Barchans Islands revealed the two-component NRM. The normal polarity A-component with the blocking temperatures 150–350 °C has $D = 334.7^\circ$ and $I = -81.4^\circ$. The high-temperature B-component is reversed with $D = 151.1^\circ$, $I = 81.8^\circ$. This direction is close to values found by Blundell (1962) from 12 quartz-diorite samples on the same area ($D = 144^\circ$; $I = 82^\circ$). The gabbros from Anagram Islands (Nob Island) have a single stable R-polarity NRM (Fig. 2i) with $D = 176.8^\circ$ and $I = 77.5^\circ$, which is close to a high-temperature B-component determined in the Barchans granodiorites.

Wiencke Island (Port Lockroy). Diorites, quartz-diorites and granodiorites were sampled in the western part of Wiencke Island located 50 km to northeast from the Argentine Islands (Fig. 1c). The K–Ar radiometric ages from quartz-diorite (biotite and hornblende) from Goudier Island, Port Lockroy, fall in time interval 48–51 Ma (Table 1; Rex 1976). The sampling was performed along the sea coast of the promontory opposite to the old British base Port Lockroy. Only univectorial R-polarity components with mean direction $D = 219.5^\circ$, $I = 84.3^\circ$ were found (Fig. 2j).

2.3 Summary of palaeodirections

A summary of site coordinates and site-mean directions, with the associated statistical parameters for all investigated sites of gabbros and granodiorite are presented in Fig. 3 and are listed in Table 2.

According to the radiometric ages, the rocks sampled along the AP coast belong to the CNS. All the samples are characterized by single component of magnetization clearly observed after removing of a small soft component, which is likely of the viscous origin. The mean site palaeodirections (Table 2, Fig. 3a) agree well with the data reported previously (Blundell 1962; Grunow 1993). Thus, it is likely that the primary NRMs of these rocks correspond to the time interval 117–84 Ma. The mean palaeomagnetic direction from these sites ($N = 9$, $D = 25^\circ$, $I = -73^\circ$, Table 3) is slightly different in declination in comparison with those reported by Grunow (1993) for the mean direction obtained from six sites (Cape Tuxen and Deliverance Point, $D = 4.7^\circ$, $I = -75.2^\circ$) and by Poblete *et al.* (2011), obtained from igneous rocks of the same ages from Gerlache Strait area ($D = 3.3^\circ$, $I = -74.9^\circ$). As is evident from Fig. 3(a), declinations of sites eight and nine are rotated clockwise in relation to those of the main group. As was mentioned earlier, this may be the result of local tectonics.

Another problematic site is the Moot Point, which is characterized by a large α_{95} circle ($\alpha_{95} = 14.3^\circ$). The results from these sites can be considered as poorly constrained and should be removed from the further analysis. In so doing, the remaining six sites (1, 2, 3, 4, 6 and 7) give an average direction of $D = 11^\circ$, $I = -76^\circ$ (Table 3), which is close to the data from Grunow (1993) and Poblete *et al.* (2011).

The mean VGP for these six sites (85°S, 190°E, Table 3), obtained for the time interval 84–117 Ma, agrees well with APWP for East Antarctica reported by Besse & Courtillot (2002), and with the Phanerozoic Synthetic Apparent Polar Wander Path (SAPW) for 100 Ma (86.1°S, 191.6°E, $\alpha_{95} = 4.2^\circ$) given by Torsvik *et al.* (2008), Fig. 3(d). Also, the mean VGP does not differ significantly from the APWP for the AP, presented by Poblete *et al.* (2011) for the age 90 Ma (Lat = – 86.8°, Long = 113.0°, $\alpha_{95} = 3.4^\circ$).

For the sites collected on the islands radiometric dating yields ages of 55–75 Ma, which are outside of the CNS, the only exception is Petermann Island. Their ChRM vectors are either of N-polarity (Berthelot Island) or R-polarity (Anagram and Barchans Islands,

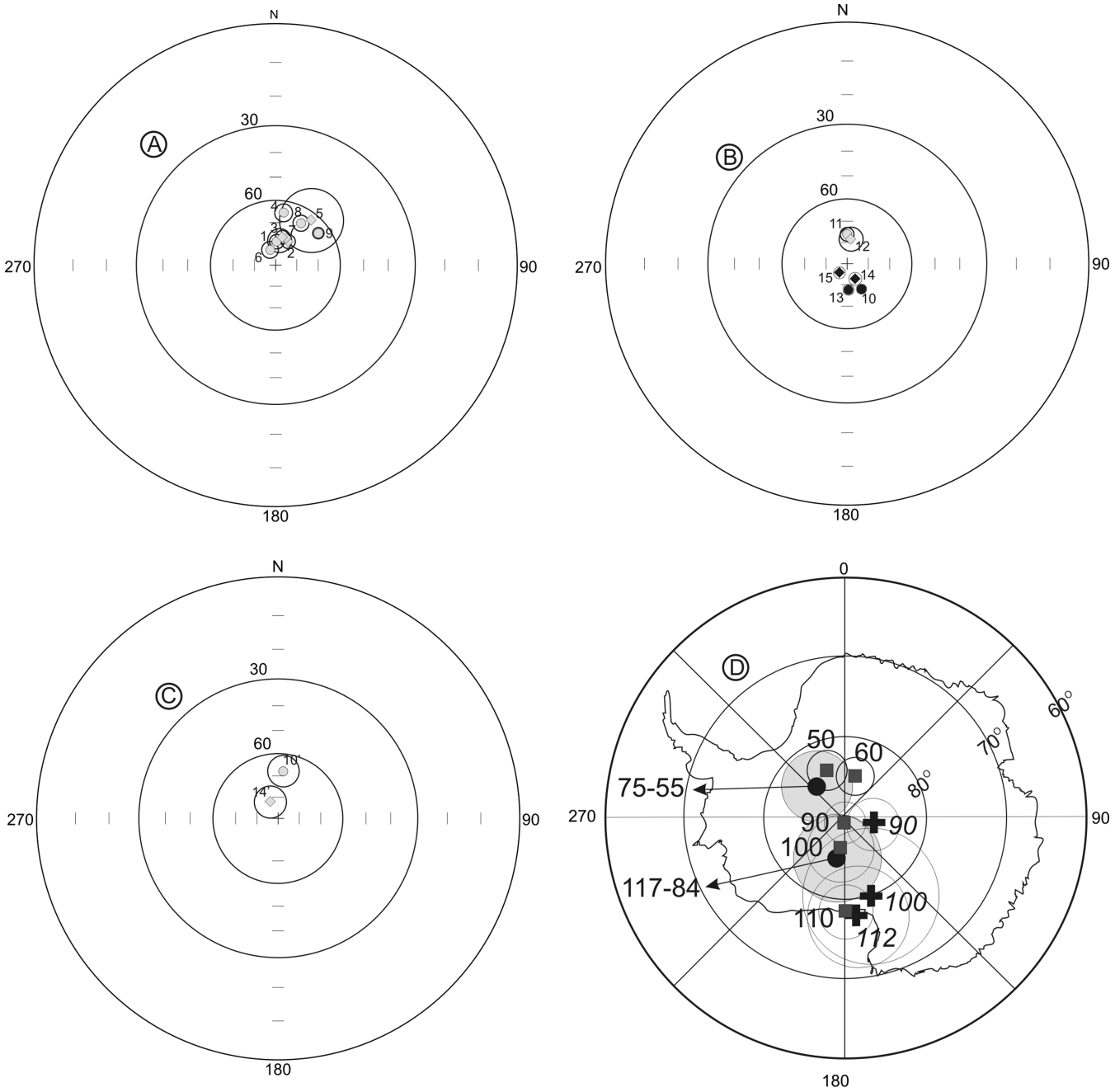


Figure 3. Site mean directions obtained with gabbro (circles) and granitoids (rhombus) rocks. The numbers near the symbols indicate the site identification number as shown in Table 2. Open (solid) symbols are plotted on the upper (lower) hemisphere by an equal area projection. (A) Directions of high temperature (450-T_c) components of NRM for sites of late Cretaceous age. (B) The same for the Palaeocene sites. (C) Directions of low temperature NRM components (T_r—450 °C). (D) Squares show reference poles with α_{95} circles for 110, 100, 90, 60 and 50 Ma from Torsvik *et al.* (2008) for East Antarctica. Crosses depict palaeopoles obtained by Poblete *et al.* (2011) for Antarctic Peninsula at 112, 100 and 90 Ma, the corresponding ages are given by italics. Black circles with shadow α_{95} ovals show our palaeopoles for 117–84 and 75–55 Ma (Table 3).

Table 3. Average mean directions and location of palaeopoles for the AP and adjacent islands.

Time interval (Ma)	Location	D°	I°	*K*	α_{95}°	*N*	Pole Lat.°S	Pole Long.° E	Pole (dp)	Pole (dm)
84–117	Antarctic Peninsula	25	–73	78	5.9	9	77	188	9	11
—	—	11	–76	148	5.5	6	85	190	9	10
55–75	Adjacent islands	174	80	237	4.4	6	85	317	8	8

Note: Overall mean directions of the gabbro and granodiorite collections and palaeopoles with the semimajor and semiminor axes of the oval of 95 per cent confidence (dm, dp). Other notations are as in Table 2.

Port Lockroy), with the mean direction $D = 174^\circ$; $I = 80^\circ$ (Fig. 3b, Table 3). This direction agrees well with Palaeocene data from northern territory (Poblete *et al.* 2011). The mean VGP for the six sites from the islands is 85°S, 317°E (Table 3), which agrees well with the SAPW for 60 Ma and 50 Ma (84.4°S, 10.3°E, $\alpha_{95} = 2.4^\circ$ and 83.8°S, 338.3°E, $\alpha_{95} = 2.5^\circ$, respectively) given by Torsvik *et al.* (2008), Fig. 3(d).

In contrast to all other sites, for three sites, Port Lockroy, Anagram Islands and Petermann Islands, positive palaeoinclinations were obtained. Ages of Anagram Islands and Port Lockroy rocks are 58 and 51 Ma, respectively. According to Gradstein *et al.* (2004), there are several intervals of reversed polarity around this age so the magnetization of rocks can be attributed to one of these intervals. The situation is more complicated when the polarity of the Petermann Island volcanic rocks is considered because of its age (93–96 Ma), it certainly refers to the CNS. Thus, the rocks are expected to have normal polarity, but the high-temperature ChRM on samples from this site is always of the reversed polarity. At the same time, the low-temperature component A, on contrary, has normal polarity. It is possible that a two-component remanent magnetization with almost opposite direction of the components can be caused by the self-reversal phenomenon. However in the case when a sample was cooling down from T_c in normal external field, the directions of the components should be of opposite order: the high-temperature component must be normally directed while the low-temperature component should be reversed. Thermal demagnetization of a laboratory induced TRMs (an example is given below, Fig. 7e2) do not favour the suggestion that rocks from Petermann Island may exhibit self-reversal as all these curves do not show any sign of self-reversal process. Hence, these observations do not support a self-reversal mechanism for the NRM of these samples. On the other hand, one can assume that the reversed sign of the high-temperature component B was caused by remagnetization of NRM at time younger than the CNS. However, it is difficult to explain why this process did not affect the low-temperature component A. In our opinion, the most likely scenario is that the high-temperature component B of Petermann Island volcanic rocks was acquired during a short event of reversed polarity inside the CNS. Indeed, a few such events are listed there (Gradstein *et al.* 2004). Taking into account that the Petermann Island structurally is a massive intrusion of sizes of a few 100 m, its cooling time can be estimated at 100 thousand years. Thus, it is possible that the high-temperature component B was acquired when a short interval of reversed polarity was present and the normal low-temperature component A was imparted at a later time when this interval of reversed polarity was already over.

Before to finish this section, for the sake of objectiveness, we must admit that an alternative interpretation of the reversed component of the Petermann Island samples may be offered. Indeed, the age for Peterman Island was obtained from granites, diorites or quartz diorites, but the palaeomagnetic study was performed on samples from gabbros. Hence, it is possible that the published ages may not reflect the age of the gabbros, which then might be not of the CNS age.

3 DETERMINATION OF PALAEOINTENSITY

3.1 Experimental procedure and rock magnetic investigation

A part of the collection (186 samples from 24 different sites) was transferred to the Geophysical Observatory 'Borok' of the Russian Academy of Sciences for palaeointensity experiments. The rock magnetic and palaeomagnetic measurements were made on a few cubic sister specimens of 1 cm in edge length cut from either drilled cores or hand samples. The Curie points, T_c, and the thermal stability of magnetic minerals were estimated from thermomagnetic heating–cooling cycles to incrementally higher temperatures T_i. These measurements were made with a Curie balance in an external magnetic field of $B = 450$ mT. To assess the magnetic hardness and mineralogy of samples, measurements of magnetic susceptibility, hysteresis loop parameters, such as coercive force B_c, remanent coercive force B_{cr}, saturation magnetization M_s and remanent saturation magnetization, M_{rs}, were performed.

The ratios M_{rs}/M_s and B_{cr}/B_c are plotted on a Day plot (Day *et al.* (1977), Fig. 4). As is seen from the diagrams, these ratios fall in either MD or PSD regions: $M_{rs}/M_s = 0.01$–0.46, $B_{cr}/B_c =$ 2–9, which suggests the presence of both relatively fine and coarse magnetic grains.

The domain structure (DS) was estimated also from the thermomagnetic criterion as defined by Shcherbakova *et al.* (2000), which estimates the DS by thermal demagnetization of the partial

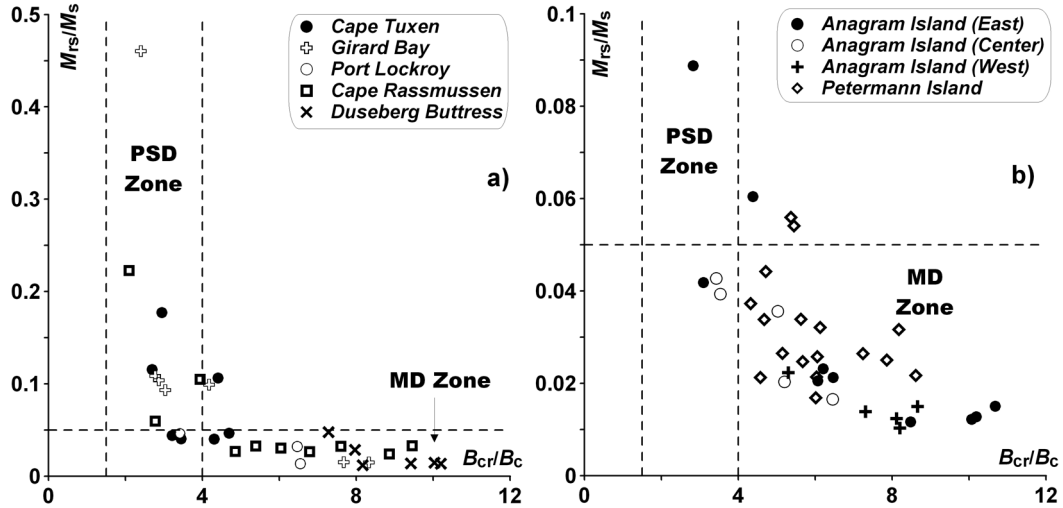


Figure 4. Day plots for samples from sites: (a) Cape Tuxen, Girard Bay, Port Lockroy, Rasmussen Island & Rasmussen Hut, Duseberg Buttress, (b) Anagram Islands, Petermann Island.

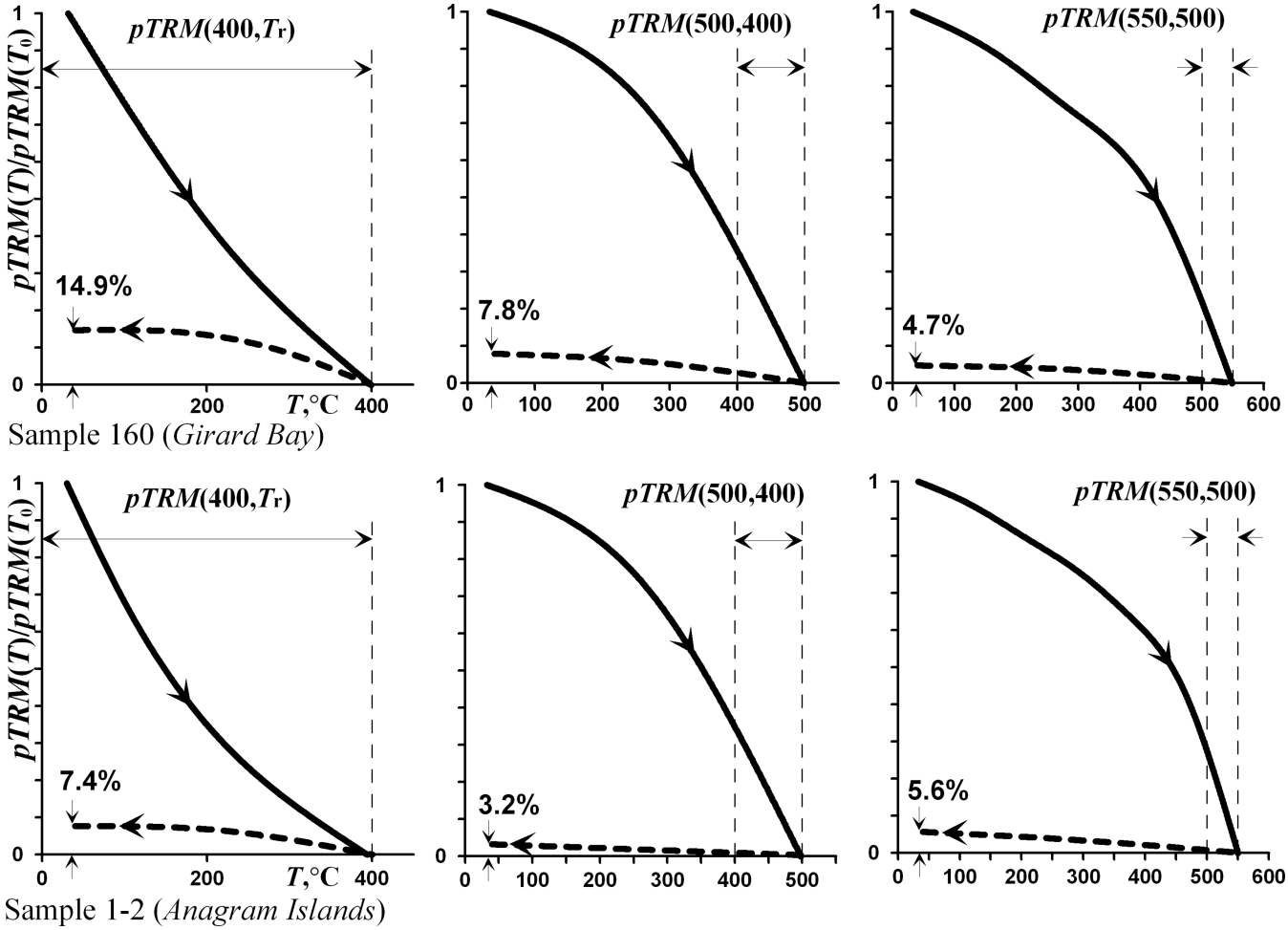


Figure 5. Application of the thermomagnetic criterion for the DS test (Shcherbakova *et al.* 2000) for sample 160, Girard Bay (upper row) and sample 1–2, Anagram Islands (lower row). Full lines are the continuous thermodemagnetization curves of the pTRMs, dashed lines are the cooling curves of the tails of the pTRMs, both heating of pTRMs and cooling of tails of pTRMs are performed in zero field. The arrows point the direction of temperature change. The corresponding pTRMs are indicated in the diagrams. Figures near the cooling curves of tails of pTRMs depict the intensity of the tails in per cents.

thermoremanent magnetization (pTRM). The criterion relies upon the Thellier law of independence of pTRMs, specifically on the equality of blocking and unblocking temperatures. The remanence remaining after the thermal demagnetization of pTRM is the so-called tail of pTRM. Typical examples of application of this test to samples from the collection are shown for sample 160 from Girard Bay (Fig. 5, upper row) and for sample 1–2 from the Anagram Islands (Fig. 5, bottom row), with pTRMs imparted in three temperature intervals: (550,500), (500,400) and (400,T_r). Surprisingly, despite the PSD-MD characteristics obtained from the Day plot (Fig. 4), small tails of pTRMs for the majority of samples were found, which indicates that the corresponding pTRMs are carried by predominantly SD-PSD grains. The most likely explanation for this discrepancy is that the particles that are the main carries of TRM belong to the small SD-PSD fraction of the total population of magnetic grains in a sample, whereas the grains responsible for the bulk hysteresis properties like M_{rs}/M_s and B_{cr}/B_c ratios, reflect the presence of coarse grains. The explanation of the high-field/remanence discrepancy was suggested by Carvallo *et al.* (2006) and Paterson *et al.* (2010). The suggestion of the co-existence of fine and coarse grains is supported by direct electron-microscopic observations as shown in Fig. 6. The examination of polished sections revealed the existence of a large ferrimagnetic grain exhibiting the exsolution structure (Fig. 6a) together with relatively small magnetite grains dispersed in the silicate matrix (Fig. 6b). Results of energy dispersive X-ray (EDX) analysis carried out for these grains for the points marked by crosses, indicates the presence of near-magnetite/near-ilmenite intergrowths (Fig. 6a) together with the presence of near-magnetite PSD size grains (Fig. 6b). The near-ilmenite lamellae seen in Fig. 6(a) often exceed 1 μm in sizes. According to experiments reported by Gapeev & Tselmovich (1983, 1986, 1992) exsolution lamellae with sizes more than 1 μm are formed as result of high-temperature oxidation of primary titanomagnetite grains taking place between 700 and 900°C. Thus, the observation of such the exsolution structures provides additional evidence that the NRM of such the grains is primary, it is of thermoremanent origin and the carriers of NRM are SD-PSD grains. The SD-PSD nature of grains-carriers of NRM and TRM is seen also from the shape of the Arai-Nagata plots (Fig. 7, column a3–i3), which exhibit straight lines. A small curvature seen in the Figs 7(a3) and (b3) can be referred to the presence of small PSD grains (Shcherbakov & Shcherbakova 2001). This suggestion is supported by Paterson (2011) who showed that zero field heating applied in the course of Thellier procedure may enhance Arai-Nagata plot curvature for small PSD-to-MD grains. The domination of PSD grains as the carriers of NRM is consistent with results of

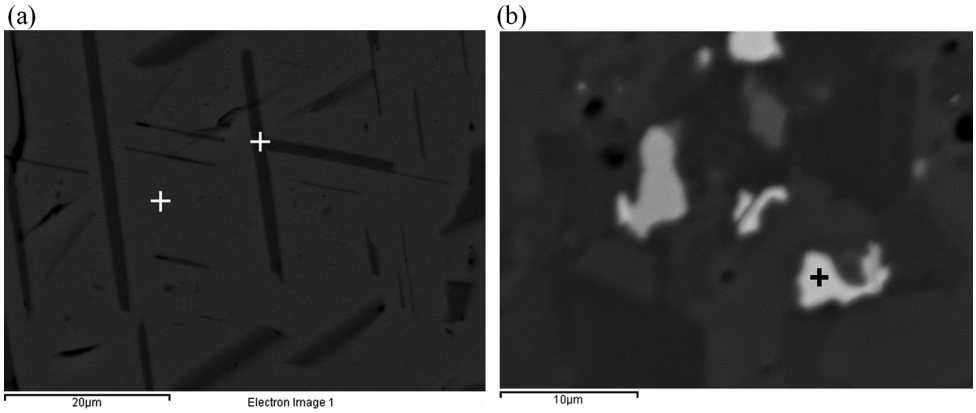


Figure 6. Photomicrographs of polished sections of samples from Cape Tuxen. (a) Sample 149. Observations of exsolution near-ilmenite lamellae (dark stripes) provide evidence for the presence of heterophase oxidation of primary titanomagnetite. (b) Sample A25, which contains PSD magnetite grains dispersed in the silicate matrix. The crosses label the sites where the energy dispersive X-ray (EDX) analysis was carried out.

the application of thermoremanent criteria, which shows only small tails of pTRMs imparted in different temperature intervals (Fig. 5). Finally, the high values of the parameter $f > 0.5$ associated with most of our intensity's determinations give a stronger line of support for a lack of MD bias (Biggin & Thomas 2003).

Palaeointensities were determined following the experimental protocol of Coe's modified Thellier–Thellier procedure (Coe 1967). The experiment consists of a sequence of paired heatings in air to a set of increasing temperature Ti, $i = 1,..n$. The first heating–cooling step to T_i takes place in zero field, the second heating is also performed in non-magnetic space following by cooling in the laboratory field, $B_{\rm lab}$, equal to 20 μT. A low external field, $B_{\rm lab}$, far less than the modern Earth's field, was deliberately chosen as pilot samples yielded palaeointensities between 10 and 20 μT and because errors are minimized when the ancient and laboratory fields are similar (e.g. Yu *et al.* 2004). Double heatings were carried out in no less than 16 steps up to 650°C, pTRM checks and susceptibility measurements were performed after every second step**.** For each sample the Arai-Nagata and Zijderveld diagrams followed to the Thellier-Coe procedure data were constructed to evaluate $B_{\rm anc}$ and to control the fit interval for a palaeointensity determination.

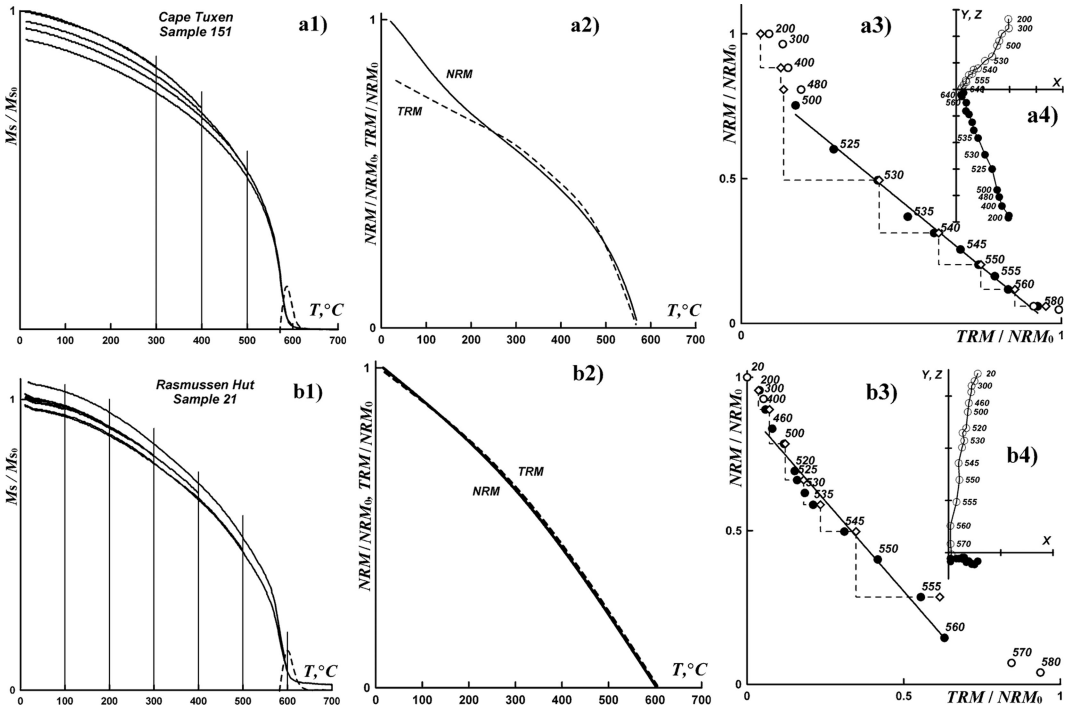


Figure 7. Strong field thermomagnetization curves $M_{\rm s}(T_{\rm i})/M_{\rm s}(T_{\rm r})$ (left-hand side), representative examples of accepted palaeointensity results of Wilson's (centre) and Thellier-Coe's experiments (right-hand side) for samples from studied sites. (a1)–(i1): Temperature dependence of saturation magnetization in $B =$ 450 mT. The set of strong field curves $M_{\rm s}(T_{\rm i})$ are obtained by heating to incrementally higher temperatures, vertical lines indicate the maximum temperatures for a given temperature loop. (a2)–(i2): Continuous thermal demagnetization curves of NRM(T) (solid lines), TRM(T) acquired in $B_{\rm lab} = 20$ μT and TRM$^* = k_{\rm w} \times$ TRM (dashed lines). All these curves are normalized to the corresponding NRM value measured at room temperature. (a3)–(i3): Arai-Nagata diagrams. (a4)–(i4): Orthogonal vector projection plots (in sample coordinates). Solid and open symbols denote horizontal and vertical projections, respectively.

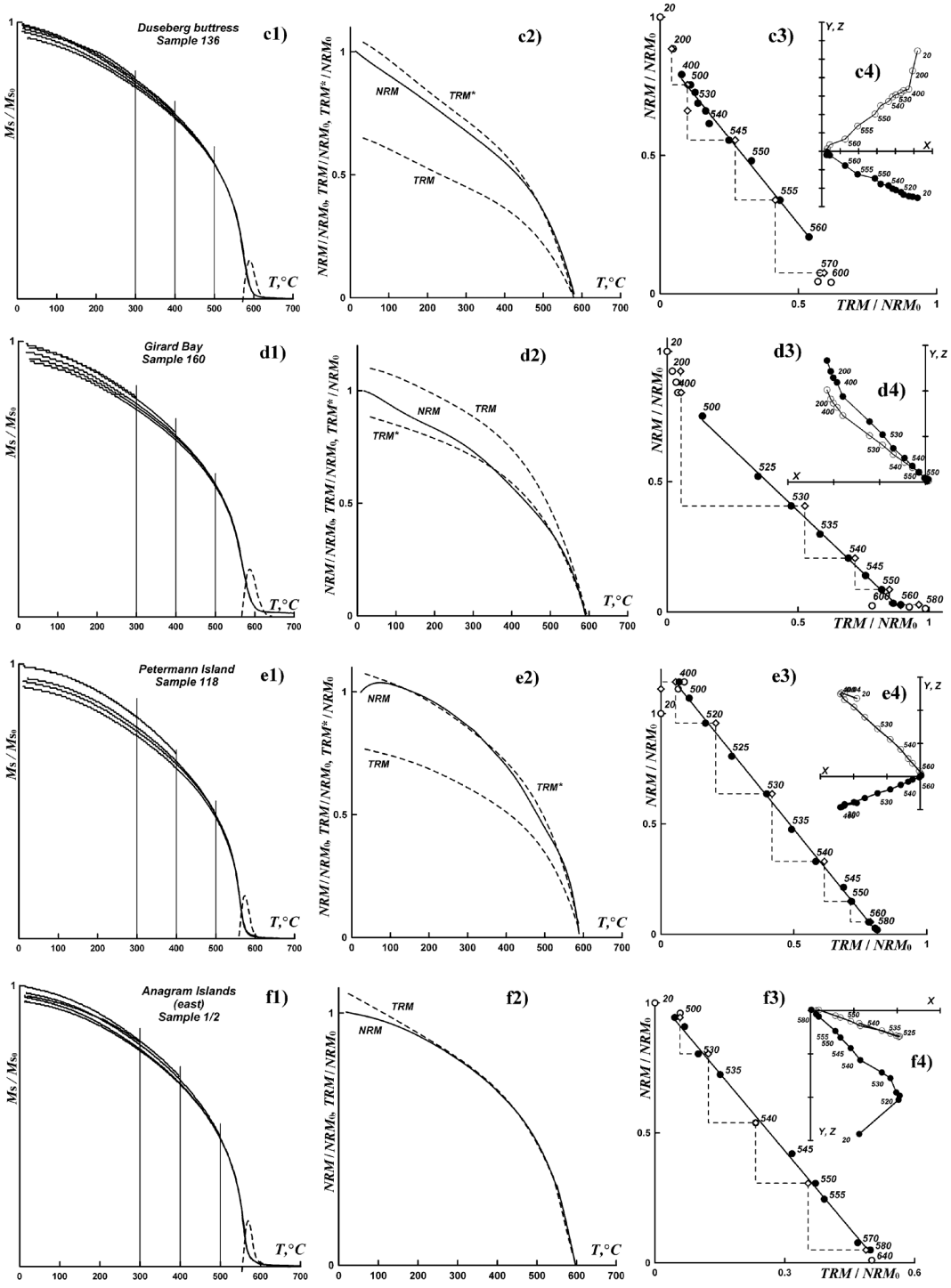


Figure 7. (*Continued.*)

To improve the level of confidence in our palaeointensity determinations, two to four sister cubes were used from each sample to obtain additional results from alternative methods. Part of the specimens were heated in an electric furnace with a residual field <50 nT, while the remaining specimens were subjected to the Thellier-Coe procedure in a full-vector three component vibrating sample magnetometer (3D-VSM) constructed at the Geophysical Observatory 'Borok', Russia. The 3D-VSM is supplied with a furnace, so it can perform a completely automated Thellier procedure. The results are recorded by a master computer that allowed us to carry out experiments on different samples using exactly the same algorithm, without extracting/placing the specimen inside the magnetometer during a Thellier experiment. The sensitivity of the 3D-VSM is 10^{-8} Am^2 and the maximum available external (vertical) field is 0.2 mT. The remanence of the specimens heated in the electric furnace was measured with JR6 magnetometer. Results of palaeointensity experiments obtained for the sister cubes, but from different devices, agree well with each other (Table 4).

© 2012 The Authors, *GJI*, **189,** 204–228
Geophysical Journal International © 2012 RAS

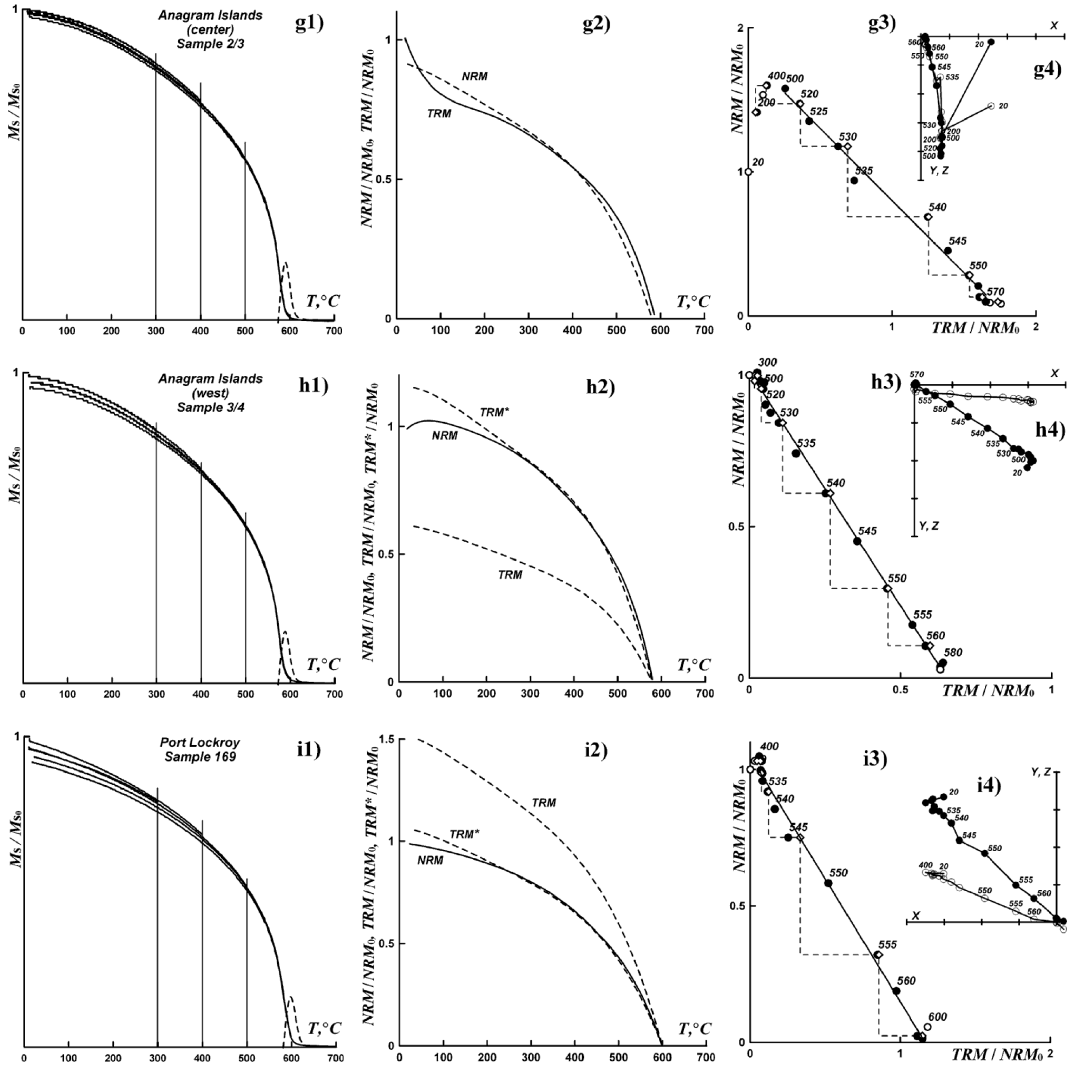


Figure 7. (*Continued.*)

In addition to the Thellier experiments, Wilson's method (Wilson 1961) of palaeointensity determinations was also applied to a number of sister specimens of those ones, which passed the selection criteria described later in the Section 3.3. The method requires a continuous record of magnetic remanence during heating to T_c and thus it was carried out using the same full-vector 3D-VSM. First the NRM is demagnetized in zero field and the thermal demagnetization curve NRM(T) is obtained. During subsequent cooling from T_c to T_r a laboratory field $B_{lab} = 20$ μT is applied to impart a TRM, which is subsequently thermally demagnetized by heating to T_c. Finally, both thermal demagnetization curves NRM(T) and TRM(T) are compared to find the temperature interval, (T_{w1}, T_{w2}), ($T_{w1} < T_{w2}$), where NRM(T) is similar to TRM(T). To perform the comparison, we introduced a fitting function TRM*(T) with a help of a multiplying coefficient k_w defining TRM*(T) = $k_w \times$ TRM(T). Then, by variation of k_w the best fit of the curves TRM*(T) and NRM(T) in a certain temperature interval (T_{w1}, T_{w2}) was judged simply by eye. Examples of results of such fits are given in Figs 7a2–i2 from which one can estimate the accuracy of the fitness as a few percents.

If such the interval (T_{w1}, T_{w2}) is found, B^*_{anc} can be obtained from the coefficient of similarity, k_w = NRM(T)/TRM(T) = B^*_{anc}/B_{lab}, $T \in (T_{w1}, T_{w2})$. Noteworthy that the similarity of NRM(T) and TRM(T) curves provides a strong argument in favour of thermoremanent nature of the NRM. Examples of thermal demagnetization curves obtained during the application of the Wilson's method are shown in Fig. 7 (column a2–i2). Note that TRM* = $k_w \times$ TRM curves are not shown in the Figs 7(a2) and (b2) because the parameter k_w for these samples is close to 1, so TRM and TRM* practically coincide.

As is seen from the Table 5, the results of the Wilson's and Thellier's methods do not contradict each other. For the better comparison, we calculated average Thellier intensities for each sample (Table 4, the column av. B_{anc}(Th)) and compared them with the Wilson determination done for this sample. As occurred, the average difference between the intensities obtained by these two methods is 2 ± 0.6 μT with the standard variation as high as 5 μT. Thus, the Wilson method produces in general slightly higher intensities and the difference is significant on 95 per cent level. Whether or not this difference has a general meaning or it occasionally changes from one to another collection, the answer to this problem requires much work, which is out of scope of this paper.

Note that in any case we do not consider here the results obtained by the Wilson's method as having the same significance as the

Table 4. Palaeointensity determinations and associated statistical parameters.

Sample	T_1–T_2 °C	N	g	q	f	B_{anc}(Th) (μT)	$\sigma_b(B_{anc})$ (μT)	B_{anc}(Th) av. (μT)	B_{anc} (Wl) (μT)	M_{rs}/M_s	B_{cr}/B_c
Cape Tuxen											
148–2_jr6	400–550	11	0.80	17.7	0.86	14.9	0.6	14.9	14.8	0.04	3.22
149	450–590	9	0.82	22.5	0.78	24.1	0.7				
149–2_jr6	400–565	13	0.88	36.8	0.79	34.1	0.6	29.1	44.0	0.04	3.45
150–2_jr6	400–550	5	0.60	7.4	0.85	20.1	1.4	20.1	19.6	0.04	4.31
151	400–580	13	0.91	16.2	0.82	20.1	0.9	20.1	19.6	0.05	4.70
A-04–04	400–565	10	0.84	12.0	0.83	20.7	1.3	20.7	–	–	–
A24–1_jr6	300–550	12	0.84	9.5	0.81	14.8	1.1	14.8	9.0	0.11	2.35
A25a	400–570	9	0.76	11.9	0.62	10.1	0.4	10.1	13.8	0.18	2.95
A26a	200–570	11	0.86	6.9	0.80	12.6	1.3	12.6	15.8	0.12	2.69
A27	400–555	11	0.87	12.1	0.73	7.3	0.4	7.3	10.8	0.11	4.41
Rasmussen Hut											
021–1	460–560	10	0.86	10.1	0.68	22.8	1.3	20.9	20.0	0.11	3.94
021a	500–570	10	0.78	5.3	0.53	19.0	1.5				
023	485–575	7	0.78	8.6	0.48	24.2	1.1				
023–2a	530–575	8	0.78	4.2	0.49	19.4	1.8	24.5	21.2	0.06	2.78
023–3a	300–565	10	0.76	3.2	0.61	29.8	4.4				
Rasmussen Island											
A28	500–600	9	0.55	14.7	0.92	31.7	1.1	30.5	29.6	0.02	8.87
A28_jr6	500–555	7	0.75	8.0	0.75	29.3	2.2				
A-29_jr6	520–560	7	0.73	26.0	0.85	31.3	0.7	31.3	23.0	0.03	7.60
A30	500–570	11	0.85	14.0	0.59	25.9	0.9	23.4	20.0	0.04	5.88
A30b	500–570	6	0.66	7.2	0.87	20.9	1.7				
A31	520–560	5	0.63	8.1	0.59	14.1	0.6	21.1	25.0	0.03	9.45
A31_jr6	520–545	4	0.50	3.6	0.76	28.1	2.9				
A32_jr6	300–545	7	0.67	6.3	0.62	25.2	1.7	24.2	34.8	0.03	4.86
A32–2_jr6	200–550	10	0.86	4.9	0.77	23.1	3.1				
A34_jr6	500–545	5	0.69	3.5	0.53	20.7	2.2	20.7	24.6	0.03	6.04
A35	420–555	10	0.85	3.0	0.35	20.1	1.9	20.1	20.6	0.22	2.10
A36	500–560	6	0.75	3.0	0.59	16.5	2.5				
A36_jr6	535–570	7	0.71	3.3	0.38	15.5	1.3	16.0	20.4	0.03	6.79
A37	500–580	8	0.69	9.2	0.56	18.9	0.8	18.9	23.2	0.03	5.39
Duseberg Buttress											
131	500–600	9	0.61	12.2	0.86	19.4	0.8	19.4	31.6	0.01	10.04
132	200–555	13	0.86	6.9	0.65	18.8	1.5	18.8	23.0	0.01	10.20
133	440–600	9	0.71	18.4	0.85	20.5	0.7	23.3	27.8	0.01	8.17
133–2	500–570	10	0.88	12.8	0.93	26.0	1.6				
134	500–580	11	0.87	19.4	0.91	40.5	1.6				
134–2_jr6	400–580	11	0.86	22.7	0.84	23.2	0.7	30.7	36.0	0.03	7.97
134–3_jr6	400–555	10	0.86	16.3	0.87	28.4	1.3				
135	300–600	11	0.79	18	0.95	19.9	0.8				
135–2	200–580	11	0.84	10.1	0.84	16.9	1.2	18.5	16.0	0.05	7.28
135–2_jr6	400–555	10	0.83	17.3	0.89	18.8	0.8				
136	400–600	14	0.87	18.5	0.75	27.5	1.0				
136–1	420–570	12	0.85	15.8	0.68	19.2	0.7	23.4	30.0	0.01	9.42
Girard Bay											
158	525–555	7	0.82	8.3	0.38	20.5	0.8				
158–2a	520–575	8	0.82	6.3	0.91	18.8	2.2	19.7	19.4	0.02	8.33
159	530–570	8	0.84	8.4	0.44	18.8	0.8				
159–2	450–560	5	0.67	20.9	0.96	17.1	0.5	17.3	19.0	0.11	2.77
159–3	530–570	8	0.84	8.4	0.44	18.8	0.8				
159a	300–565	10	0.85	22.2	0.88	14.5	0.5				
160	400–555	10	0.83	33.6	0.81	19.9	0.4	19.9	16.0	0.09	3.03
161	400–565	12	0.90	16.2	0.70	15.8	0.6				
161a	500–560	8	0.84	10.2	0.68	17.7	1.0	16.8	13.6	0.10	2.87
161–2a	500–555	7	0.83	4.5	0.57	13.3	1.4				

Table 4. (*Continued.*)

Sample	T_1–T_2 °C	N	g	q	f	B_{anc}(Th) (μT)	$\sigma_b(B_{anc})$ (μT)	B_{anc}(Th) av. (μT)	B_{anc} (Wl) (μT)	M_{rs}/M_s	B_{cr}/B_c
162	500–560	10	0.85	14.9	0.66	17.2	0.7	17.2	21.2	0.10	4.17
164	525–560	8	0.85	9.9	0.66	19.3	1.1	19.0	22.8	0.46	2.40
164a	500–555	7	0.80	3.9	0.65	18.6	2.5				
165–1	400–560	11	0.89	10.4	0.79	19.7	1.3	19.7	24.4	0.02	7.68
Petermann Island											
107 (0.08)	520–600	12	0.88	26.4	0.86	28.5	0.8				
107a (0.11)	400–570	9	0.77	13.6	0.91	22.8	1.2	25.7	22.6	0.02	6.03
108* (0.17)	520–600	11	0.89	51.2	1.08	27.4	0.5	27.4	27.2	0.03	6.06
109–1a	400–600	12	0.85	19.4	0.75	23.2	0.8				
109	450–565	8	0.79	36.6	0.92	24.5	0.5	23.9	26.0	0.03	7.86
110* (0.16)	400–640	15	0.91	35.8	0.99	33.5	0.8	33.5	21.8	0.03	4.67
111a* (0.11)	500–560	6	0.69	12.6	0.96	20.7	1.1				
111* (0.11)	450–590	8	0.583	11.064	1.12	35.1	1.7	27.9	31.8	0.03	6.13
112	400–570	10	0.738	18.25	0.956	38.5	1.9				
112–2a	520–570	7	0.778	9.854	0.878	27.9	1.4	33.2	–	0.03	8.18
113–2a* (0.07)	530–570	6	0.69	9.8	0.80	24.5	1.4	24.6	29.0	0.03	7.24
114a* (0.06)	500–560	6	0.71	13.9	0.76	26.1	1.0				
114–1* (0.09)	350–570	9	0.75	37.4	1.08	29.0	0.6	27.6	29.0	0.06	5.37
116a* (0.07)	300–550	7	0.70	15.4	0.94	21.1	0.9				
116* (0.07)	400–560	5	0.61	143.9	0.97	26.3	0.1	23.7	32.0	0.04	4.71
118* (0.15)	400–580	12	0.89	79.4	0.98	29.8	0.3				
118–1* (0.04)	440–590	11	0.73	20.5	1.02	25.5	0.9	27.7	27.6	0.03	5.15
119	400–555	9	0.86	39.9	0.84	25.6	0.5				
119a-1* (0.07)	500–590	7	0.70	43.3(?)	0.97	20.5	0.3	23.1	31.2	0.02	8.61
120a* (0.16)	400–590	11	0.75	15.8	0.90	20.0	0.9				
120–1* (0.17)	350–575	7	0.59	35.1(?)	1.18	29.0	0.6	24.5	23.6	0.04	4.32
A39–2* (0.04)	530–580	9	0.84	17.3	0.89	18.3	0.8	24.6	20.0	0.03	5.63
A41* (0.2)	300–580	13	0.85	41.1	1.23	23.2	0.6				
A41–2a* (0.16)	300–580	12	0.81	12.7	0.92	22.4	1.3	20.8	23.0	0.02	4.58
A41a* (0.17)	400–560	8	0.76	7.6	0.73	16.8	1.2				
A42* (0.06)	400–600	15	0.91	37.2	0.98	38.1	0.9				
A42–2	500–600	13	0.89	23.0	0.88	33.0	1.1				
A42a	200–600	14	0.85	35.1	0.91	21.5	0.5	30.9	31.0	0.02	5.67
A43* (0.18)	400–600	15	0.89	41.6	0.94	24.6	0.5	24.6	21.0	0.05	5.45
Anagram Islands (east)											
1–1	300–570	13	0.88	40.9	0.84	39.8	0.7	39.8	44.8	0.04	3.09
1–2	500–580	11	0.86	36.5	0.91	40.4	0.9				
1–2a	400–575	11	0.85	15.4	0.97	41.3	2.2	40.9	39.2	0.02	8.16
1–3	20–600	14	0.89	27.5	0.98	43.5	1.4				
1–3a	300–575	12	0.82	17.6	0.90	33.5	1.4	38.5	46.4	0.02	6.47
1–4	500–570	10	0.80	10.7	0.81	44.9	2.7				
1–4a	300–575	12	0.83	21.8	0.90	32.1	1.1	38.5	49.2	0.02	10.68
1–5	500–600	13	0.90	40.3	0.96	46.3	1.0				
1–5a	500–595	12	0.83	11.6	0.87	42.0	2.6	44.2	48.0	0.02	6.07
A51	400–555	6	0.48	13.2	0.97	37.7	1.3				
A51–2	500–580	11	0.76	15.1	0.96	42.1	2.0	40.0	–	0.06	4.38
A51a	400–595	11	0.84	12.7	0.96	40.3	2.6				
A52	500–580	11	0.79	23.5	0.97	35.9	1.2				
A52–2a	520–575	9	0.83	26.7	0.99	25.6	0.8	30.0	–	0.01	8.48
A52a	500–560	7	0.66	9.5	0.87	28.5	1.7				
A54–2	500–580	11	0.83	28.6	0.99	41.7	1.2	41.7	40.2	0.01	10.19
A55a	520–570	8	0.77	17.7	0.99	25.5	1.1	25.5	35.8	0.02	12.88
A56–2	520–580	8	0.80	21.7	0.59	28.5	0.6	28.5	–	0.09	2.83
A57	400–580	13	0.91	24.9	0.93	32.6	1.1	32.6	34.8	0.01	10.07
Anagram Islands (centre)											
2–1	500–570	11	0.86	47.3	0.99	27.7	0.5				
2–1-2a	520–575	9	0.81	24.9	0.91	22.8	0.7	25.3	32.8	0.02	6.45
2–2	500–540	5	0.59	6.0	0.72	22.9	1.6	19.8	20.0	0.04	3.43

Table 4. (*Continued.*)

Sample	T_1–T_2 °C	N	g	q	f	B_{anc}(Th) (μT)	$\sigma_b(B_{anc})$ (μT)	B_{anc}(Th) av. (μT)	B_{anc} (Wl) (μT)	M_{rs}/M_s	B_{cr}/B_c
2–2a	400–560	9	0.85	11.7	0.84	16.6	1.0				
2–3	500–580	12	0.90	25.0	0.76	19.2	0.5	20.4	20.8	0.04	3.53
2–3-3a	20–540	8	0.79	4.1	0.47	20.4	1.8				
2–3a	400–545	6	0.71	6.9	0.71	21.7	1.9				
2–4	500–580	12	0.89	26	1.0	25.6	0.9	23.5	29.0	0.02	5.20
2–4a	530–560	6	0.76	8.6	0.81	21.4	1.5				
2–5	500–570	11	0.80	19.2	0.58	42.5	1.0	39.7	31.4	0.04	5.03
2–5-2	400–560	11	0.84	17.1	0.91	36.8	0.3				
Anagram Islands (west)											
3–2	500–580	11	0.87	18.2	0.92	25.3	1.1				
3–2-2	500–580	11	0.83	42.3	0.94	27.2	0.5	24.7	24.6	0.02	5.30
3–2a	520–570	9	0.84	20.7	0.98	21.7	0.9				
3–3	500–570	11	0.86	36.6	0.98	24.9	0.6				
3–3a	530–575	8	0.81	14.2	0.97	23.1	1.3	24.0	28.8	0.01	8.11
3–4	300–580	15	0.88	48.4	0.96	31.0	0.5				
3–4a	500–570	10	0.82	50.7	0.92	30.3	0.5	30.7	38.0	0.02	8.66
3–5	500–570	11	0.81	20.1	0.99	40.7	1.6				
3–5a	530–570	8	0.80	11.7	0.85	30.8	1.8	35.8	40.4	0.01	8.20
3–6-2	300–560	13	0.87	93.1	0.94	40.0	0.4				
3–6-2a	530–575	8	0.83	9.4	0.84	29.2	2.3	34.6	28.4	0.01	7.31
Port Lockroy											
167	520–570	10	0.85	24.1	0.92	19.3	0.6				
167–2	520–580	4	0.61	130.3	0.72	17.0	0.1	17.2	21.2	0.05	3.42
167–2_jr6	500–545	7	0.70	6.7	0.78	15.3	1.2				
168	400–565	8	0.73	17.4	0.66	18.8	0.5				
168–2_jr6	400–565	11	0.86	14.0	0.93	14.8	0.8	17.7	19.0	0.01	6.55
168a	400–565	9	0.63	3.5	0.61	19.6	2.2				
169	400–580	13	0.85	31.7	1.03	18.1	0.5				
169–2a	400–575	11	0.82	18.6	0.76	10.3	0.3	14.8	14.2	0.03	6.47
169–2_jr6	400–540	7	0.69	7.3	0.71	16.0	1.1				
176	500–580	12	0.83	14.5	0.88	22.3	1.1	24.2	34.0	0.01	5.81
176–3	400–570	11	0.85	18.8	0.88	26.0	1.0				

Note: The column (T_1, T_2) designates the temperature interval of the Arai–Nagata diagram used for the approximation of the B_{anc} values, N is the number of points within this interval used to determine the best linear fit. The notations $\boldsymbol{B}_{anc}$**(Th),** $\boldsymbol{B}_{anc}$(Wl) are the palaeointensity estimates obtained by the Thellier-Coe's and the Wilson's methods, respectively, $\sigma_b(B_{anc})$ is the standard error of determination of $\boldsymbol{B}_{anc}$(Th), $\boldsymbol{B}_{anc}$(Th) av. is the average palaeointensity calculated for the given sample. The parameters f, g, σ and q characterize the quality of the results obtained (Coe *et al.* 1978): f represents the fraction of NRM spanned by the linear fit; the gap factor g quantifies the uniformity of the distribution of successive data points in a chosen temperature interval, the quality factor $q = k \times f \times g/\sigma(k)$ quantifies the general quality of a given palaeointensity determination with the coefficient $k = B_{anc}(\text{Th})/B_{lab}$ being the slope of the linear fit to the Arai–Nagata plot and $\sigma(k)$ being its standard error. The last two columns present the hysteresis parameters M_{rs}/M_s and B_{cr}/B_c values to the entries.

Samples marked as 'jr6' and 'a' were processed in a small furnace, the remaining samples were heated in the 3D-VSM. The majority of samples from Petermann Island have two-component NRM, with almost antiparallel directions. These samples are marked with asterisks, the figure inside the brackets after the asterisk evaluates the fraction of low temperature components of the NRMs in relation to the value of the corresponding high temperature component. As the contribution of the low temperature component was not taken into account when the value of the total NRM was computed, the parameters f and q for the marked samples are slightly overestimated.

Thellier results. Indeed, because of the informal way of the estimation of the fitting parameter k_w, the results from the Wilson's method can be treated only as providing auxiliary information helping to select the Thellier palaeointensity determinations. Though we must admit that recently Muxworthy (2010) suggested formalizing the Wilson's procedure by the analogy with the Thellier method. For this, he proposed to estimate the slope of the best-fit NRM-TRM segment by the same way as it is routinely done for the Thellier method. This way can in principle bring a new status to the Wilson's method if it will be well developed experimentally and methodically by future investigations.

3.2 Magnetic properties

The magnetic properties of samples selected for the final palaeointensity data set are similar. Thermomagnetic curves, $M_s(T)$, only differ slightly from each other and exhibit good thermal stability of the ferrimagnetic minerals with Curie temperatures 570–590°C (Figs 7, column a1–i1). These date indicate that the mineral is magnetite or low titanium titanomagnetite in agreement with the electron-microscopic observations (Fig. 6). A small progressive decrease of $M_s(T_r)$ intensity with heating steps, which totals no more than 10–15 per cent of the total NRM, is most likely related

Table 5. Mean palaeointensities (B_{anc}) and virtual dipole moments (VDM) obtained by the Wilson and Thellier-Coe methods and associated VDMs.

		Wilson method			Thellier-Coe method					
Site	Age (Ma)	n_{WL}	B_{anc} (μT)	$\sigma(B_{anc})$ (μT)	n/n_{TL}	B_{anc} (μT)	$\sigma(B_{anc})$ (μT)	I_{anc},$^\circ$	VDM* ($\times 10^{22}$ Am^2)	VDM + SD ($\times 10^{22}$ Am^2)
Cape Tuxen	88.1 $\pm$1.1	8	18.4	11.0	9 /10	17.9	7.7	$-$79.1	2.4	2.3 $\pm$1.0
Rasmussen Hut & Rasmussen Island	117.0 $\pm$0.8	11	23.9	4.6	11/19	23.0	5.3	$-$78.9	3.1	2.9 $\pm$0.7
Duseberg Buttress	105.7 $\pm$0.7	6	27.4	7.0	6 /12	23.3	6.6	$-$67.2	3.6	3.4 $\pm$1.0
Girard Bay	88.1–96.0	7	19.5	3.8	7 /14	17.9	2.1	$-$65.1	2.9	2.7 $\pm$0.3
Petermann Island	96.0 $\pm$1.0	15	26.5	4.2	17/29	26.0	5.6	76.0	3.7	2.9 $\pm$0.6
Anagram Islands	58.0 $\pm$1.0	18	35.1	8.9	21/41	31.8	8.4	77.5	4.4	3.5 $\pm$0.9
Port Lockroy	50.0 $\pm$1.5	4	22.1	8.5	4 /11	18.0	4.1	84.3	2.4	1.9 $\pm$0.4

Note: n_{WL} is the number of samples used for the Wilson's method; n/n_{TL} is the number of successful samples/the number of successful sister cubes used for palaeointensity determination by the Thellier-Coe method; σ is the standard deviation; VDM and VDM* are the virtual dipole moments before and after the correction for cooling rate, respectively; I_{anc} is the site-mean palaeoinclination. The mean VDM values are calculated only from the results of Thellier-Coe method. SD here is the standard deviation.

to the loss of magnetite due to its transformation to maghemite and/or hematite. We note that all samples demonstrate a good similarity between the NRM(T) and TRM(T) thermomagnetic curves over a wide temperature range (Figs 7, column a2–i2), which confirms the thermal stability of the ferrimagnetic minerals within the samples and indicates the thermoremanent nature of NRMs.

3.3 Results of the palaeointensity determinations

A total of 334 sister cubes from 186 different rock samples were used in the palaeointensity experiments and satisfactory determinations were obtained from 75 samples (139 sister cubes) from seven sites: Cape Tuxen, Rasmussen Island and Rasmussen Hut, Duseberg Buttress, Girard Bay, Petermann Island, Anagrams Islands and Port Lockroy. Examples of Arai-Nagata diagrams and corresponding Zijderveld plots are shown in Fig. 7 column (a3)–(i3) and column (a4)–(i4), respectively.

A palaeointensity result for a sample was accepted only if it satisfied the following selection criteria set (Heunemann *et al.* 2004; Shcherbakova *et al.* 2008):

(1) There exists a wide enough temperature interval (T_{w1}, T_{w2}), ($T_{w1} < T_{w2}$), not less than the temperature interval (T_1, T_2), used for the linear fit on the Arai-Nagata diagram, over which the curves NRM(T) and TRM(T) are similar (positive outcome of the Wilson's method);

(2) The linear fit on the Arai-Nagata diagram used for palaeointensity determination should contain at least four consecutive data points over a temperature interval (T_1, T_2), ($T_1 < T_2$);

(3) The NRM vector is univectorial through the temperature range (T_1,T_2) used for palaeointensity determination and the best fit direction should go through the origin of the orthogonal plot;

(4) The NRM fraction f used in the linear fit represents at least 20 per cent of the total NRM;

(5) The difference between initial pTRM acquisition and pTRM check normalized to the total NRM must be less than 5 per cent within the total (T_1, T_r) interval;

(6) The susceptibility remains constant within 10 per cent in the (T_1, T_r) range.

Table 4 summarizes the results of the palaeointensity determinations that passed the selection criteria. A full description of the table is given in the table caption.

According to Coe *et al.* (1978), reliable data should have the quality factor $q \geq 5$ and one can see from Table 4 that almost all samples satisfy this criterion.

3.4 Cooling rate correction

The palaeointensity data presented in Table 4 are not corrected for the difference in cooling rates between laboratory and natural cooling. According to Dodson & McClelland-Brown (1980), the intensity of magnetization should increase by a few percent for each order of magnitude increase in cooling time. The samples from Girard Bay, Cape Tuxen, Rasmussen Island and Duseberg Buttress were taken from dykes of 1–2 m in size, so they are likely to have cooled down from T_c to about 400°C rapidly, in a few days. It is likely that in this case palaeointensities obtained without the cooling rate correction overestimate the true values by a few percent. The sampling from Petermann Island, the Anagram Islands and Port Lockroy was performed from massive intrusions hundreds of meters in size. In these cases cooling may have taken tens of thousands of years and the results of the Thellier experiments carried out for these sites may require a substantial correction.

A comprehensive theoretical study of process of blocking of magnetic moment, m, of SD grain in different cooling rates was done by Dodson & McClelland-Brown (1980). However, their formulation is not transparent and convenient for quick estimations. Later we present a simple approximation that can be easily applied to an SD sample provided that some hysteresis parameters of the grains are known.

As known from Néel (1955), for an ensemble of identical SD grains

$$\mathrm{TRM} = m(T_r)n \tanh \frac{m(T_b)h}{kT_b} \approx m^2(T_r)n\frac{j_s(T_b)h}{kT_b}, \quad (1)$$

where n is the number of grains in volume unit, T_b is the blocking temperature, k is the Boltzmann's constant and h is the week applied field. Hence,

$$\frac{\mathrm{NRM}}{\mathrm{TRM}} = \frac{j_s(T_{b.nat})T_{b.lab}}{j_s(T_{b.lab})T_{b.nat}} = \frac{j_s(T_{b.nat})(1-\tau_{b.nat})}{j_s(T_{b.lab})(1-\tau_{b.lab})}, \quad (2)$$

where $j_s(T) = M_s(T)/M_s(0)$ is the reduced spontaneous magnetization, $T_{b.nat}$ and $T_{b.lab}$ are the blocking temperatures at laboratory and natural conditions, respectively. The parameter

$$\tau = (T_c - T)/T_c, \quad (3)$$

is the reduced temperature, $\tau_{b.nat}$ and $\tau_{b.nat}$ are the reduced blocking temperatures at laboratory and natural conditions, respectively. The

starting point for the analysis is the Nell's principle that at a given blocking temperature, the relaxation time

$$t_r = \frac{1}{f_0}\exp\left(\frac{\varepsilon}{kT}\right) \quad (4)$$

is of the same order as the effective cooling time θ. Here f_0 is the characteristic frequency of attempts to overcome the potential barrier ε. From here, the blocking temperature, T_b, can be approximately found as the root of the transcendental equation $\varepsilon(T_b)/kT_b = \ln[f_0\theta(T)]$.

More precise analysis of the kinetics of the blocking process done by Dodson & McClelland-Brown (1980), led to a slightly modified expression

$$t_r = \frac{1}{f_0}\exp\left(1.78\frac{\varepsilon}{kT}\right). \quad (5)$$

To determine θ, Dodson & McClelland-Brown (1980), note that in the neighbourhood of the blocking temperature we can write $\varepsilon/kT = a_0 + t/\theta$. This equation can be rewritten as

$$\theta = 1/\frac{d}{dt}\left(\frac{\varepsilon}{kT}\right). \quad (6)$$

Let us assume for the cooling law $T(t)$ an exponential decay,

$$T(t) = T_c\exp\left(-\frac{t}{t_{cool}}\right). \quad (7)$$

The temperature behaviour of the potential barrier $\varepsilon(T)$ and the reduced spontaneous magnetization $j_s(T)$ can be modelled by widely used approximations

$$\varepsilon(T) = \varepsilon_0 j_s^r(T), \quad (8)$$

$$j_s(T) = [(T_c - T)/T_c]^\alpha, \quad (9)$$

where ε_0, r and α are empirical parameters. The parameter α can be identified as the critical exponent. In the frame of mean field approximation $\alpha = 0.5$, but thermal fluctuations usually decrease this value. It should be noted that in strictest sense the concept of critical exponent is applicable only in vicinity of T_c, but in practical applications eq. (9) is used in a wide temperature interval. Now, accounting for (8) and (9), eq. (6) can be written as

$$\frac{1}{\theta} = \frac{d}{dt}\left(\frac{\varepsilon}{kT}\right) = \frac{\varepsilon_0}{t_{cool}kT_c}[\tau^{r\alpha-1}(1+\tau)]. \quad (10)$$

Subsequently, eq. (5) becomes

$$\frac{\varepsilon_0 j_s^r}{kT_c(1-\tau)} = \ln\left[\frac{1.78 f_0 t_{cool}}{\frac{\varepsilon_0}{kT_c}[\tau^{r\alpha-1}(1+\tau)]}\right]. \quad (11)$$

Let t_{lab} and t_{nat} are the cooling times (t_{cool}) when the specimen is cooled in a laboratory or in nature, respectively. For small τ ($\tau << 1$), we can neglect difference in $1 - \tau_{b,lab}$ and $1 - \tau_{b,nat}$ in eq. (2) and τ in comparison with one in (11). This allows us to derive a simple analytical estimation where eq. (2), combined with eq. (11), can be reduced to

$$\frac{NRM}{TRM} \cong \frac{j_s(T_{b.lab})}{j_s(T_{b.nat})} = \left(\frac{A + \ln t_{lab}}{A + \ln t_{nat}}\right)^{1/r} \approx 1 - \frac{1}{rA}\log_{10}\frac{t_{nat}}{t_{lab}}, \quad (12)$$

where the parameter

$$A = \log_{10}\left[\frac{kT_c}{\varepsilon_0}\frac{1.78 f_0}{\tau^{r\alpha-1}}\right]. \quad (13)$$

Using the estimated values $f_0 \sim 10^{10}$ s^{-1}, $r \approx 3$, $\alpha \approx 0.5$, $\tau \sim 0.1$ and $kT_c/\varepsilon_0 \sim 0.01$, we get from (13) $A \approx 10$. Then from (12) we obtain that the ratio NRM/TRM increases by about 3 per cent for each order of value increase of the t_{nat} over t_{lab} in general accordance with the previous estimations (Dodson & McClelland-Brown 1980; Halgedahl *et al.* 1980).

To check the validity of (12) and (13), we carried out experiments on TRM intensity dependence on cooling rate with sample 131 from Duseberg Buttress. According to the thermomagnetic criteria, the magnetic carriers of this sample are mainly small PSD grains. The sample was first thermostabilized by annealing at 600 °C for 1 hr and then three regimes of cooling rate $1/t_{cool} = 3$, 0.1 and 0.01 deg s^{-1} were applied using the above described full-vector 3D-VSM. The averaged TRM intensities for four successive sets of these experiments (with the three regimes of cooling rate) are: 2.21 A m^{-1}, 2.28 A m^{-1} and 2.32 A m^{-1}, respectively, with the error of determination 0.02 A m^{-1}. In other words, the TRM intensity increases by about 2 per cent for an order of magnitude increase of the cooling rate. The measurements of temperature behaviour of the coercive force B_c and spontaneous magnetization M_s carried out in Borok on the VFTB magnetometer allowed us to estimate the parameters $\alpha \approx 0.35$ and $r \approx 4$. The last estimate was obtained with help of well-known relationship $\varepsilon \propto M_s H_c$. Blocking temperature range for this sample lies between T_c and 500°C, what corresponds to $\tau \sim 0.01$–0.1. Putting these values in (11) and (12), we indeed obtain that the TRM intensity should increase about (2–3 per cent) for an order of value of the cooling rate. Similar results were obtained for the sample 167 taken from the slow cooling intrusive rock from Port Lockroy. For this sample the averaged TRM intensities for four successive sets of the same three regimes of cooling rate were: 8.02 A m^{-1}, 8.37 A m^{-1} and 8.52 A m^{-1}, respectively, with the error of determination 0.04 A m^{-1} giving again about 2–3 for an order of value of the cooling rate.

Based on the above results, palaeointensities determined from Girard Bay, Rasmussen Hut and Rasmussen Island, Cape Tuxen and Duseberg Buttress are likely overestimated by about 6 per cent. The data for Petermann Island, the Anagram Islands and Port Lockroy sites, however, may be overestimated by as much as 20 per cent; their characteristic cooling time $\sim 10^{10}$–10^{12} s, which exceeds t_{lab} by 8–10 orders of magnitude. It should be noted that this correction must be regarded as the upper limit because TRM carried by PSD and MD grains may not have a dependence on cooling rate (McClelland-Brown 1984; Winklhofer *et al.* 1997). Hence, the presence of such the grains can substantially decrease the difference in TRM intensities for different cooling rate.

4 DISCUSSION

The site average palaeointensity determinations are summarized in Table 5. The VDMs are given both before and after the cooling rate correction, the notations are VDM and VDM∗, respectively. In accordance with the previous section, the correction for Girard Bay, Rasmussen Hut and Rasmussen Island, Cape Tuxen and Duseberg Buttress was taken to be 6 per cent and for Petermann Island, the Anagram Islands and Port Lockroy it was estimated as 20 per cent.

The age of Cape Tuxen rocks is taken according to Gladkochub *et al.* (2009) as this is the most recent determination carried out on the rocks studied here. There are two-age determination for the Rasmussen Island (Table 1). The U–Pb zircon age determination from coarse-grained pink granite on Rasmussen Island yielded three concordant fractions with an age of 117 ± 0.8 Ma (Tangeman *et al.* 1996), which differs markedly from the three-point Rb–Sr isochron 128 ± 3 Ma (Pankhurst 1982). For this work we use the more recent

and reliable Tangeman *et al.* (1996) dating. As was noted in the Section 2.2.1, no ages are known for Girard Bay and Duseberg Buttress, but the similarity of geological structures between these sites and the Moot Point, Petermann Island and Cape Tuxen sites allowed us to estimate the ages of Girard Bay and Duseberg Buttress given in Table 5 as combinations of the ages of Petermann Island, Moot Pont and Cape Tuxen.

All palaeointensity determinations reported in Table 5 indicate a low dipole moment in comparison with the present day VDM $\approx 8 \times 10^{22}$ Am^2.

However, the possibility that VDMs may be linked to the thermochemical remanence TCRM rather than to the TRM (Smirnov & Tarduno 2005) should be discussed. For this, below we list the evidence supporting that ChRMs studied in this paper are primary TRMs. First note that the samples accepted for the palaeointensity determinations, are characterized by single component ChRM and the mean site palaeodirections agree well with the data reported previously (Blundell 1962; Grunow 1993). From here we can conclude that the primary NRMs of these rocks correspond to the CNS time interval and were not remagnetized later. Thus, if the TCRM was imparted, it happened during the CNS.

Secondly, strong field thermomagnetic curves $M_{si}(T)$ (Fig. 7, column a1–i1) exhibit good thermal stability, hence, magnetic minerals in the samples selected for the palaeointensity studies are stable against heating. Thirdly, the similarity of NRM(T) and TRM(T) curves provides strong argument in favour of a thermoremanent nature of NRMs and their primary origin. Fourthly, important observations in favour of thermoremanent nature of NRMs are given also by results of direct electron-microscopic observations and EDX analysis as the examination of polished sections revealed the presence of PSD grains both as individual grains and as a result of the high-temperature exsolution of originally MD titanomagnetite grains. A more detailed discussion on this subject is given in Shcherbakova *et al.* (2008).

To review the statistics of VDM values for the periods studied in this paper, a compilation of Palaeocene and Cretaceous classical Thellier-type palaeointensity data from the world palaeointensity databases http://earth.liv.ac.uk/pint/ (Biggin *et al.* 2010) and http://wwwbrk.adm.yar.ru/palmag/index.html was carried out. These databases are the updated to the present day IAGA World palaeointensity database described by Perrin & Shcherbakov (1997) and Perrin & Schnepp (2004). Results of the compilation, presented in Fig. 8, reveal a number of high geomagnetic intensities for the CNS (e.g. data from Constable *et al.* 1998; Tarduno *et al.* 2006; Tauxe 2006). The Cretaceous VDMs range over a wide interval (1.1–14) $\times 10^{22}$ Am^2, with the mean VDM $= 5.5 \times 10^{22}$ Am^2. A similar remark for the CNS was recently expressed by Zhu *et al.* (2008): 'high quality palaeointensity data demonstrate that the field intensity dramatically fluctuated during the superchron'.

A motivation to perform the compilation of Cretaceous Thellier-type palaeointensity data is the search for possible relationships between the reversal rate and average VDM intensity, which was hypothesized by Cox (1968) and has been a matter of extensive discussion over recent decades. Based on Thellier palaeointensity

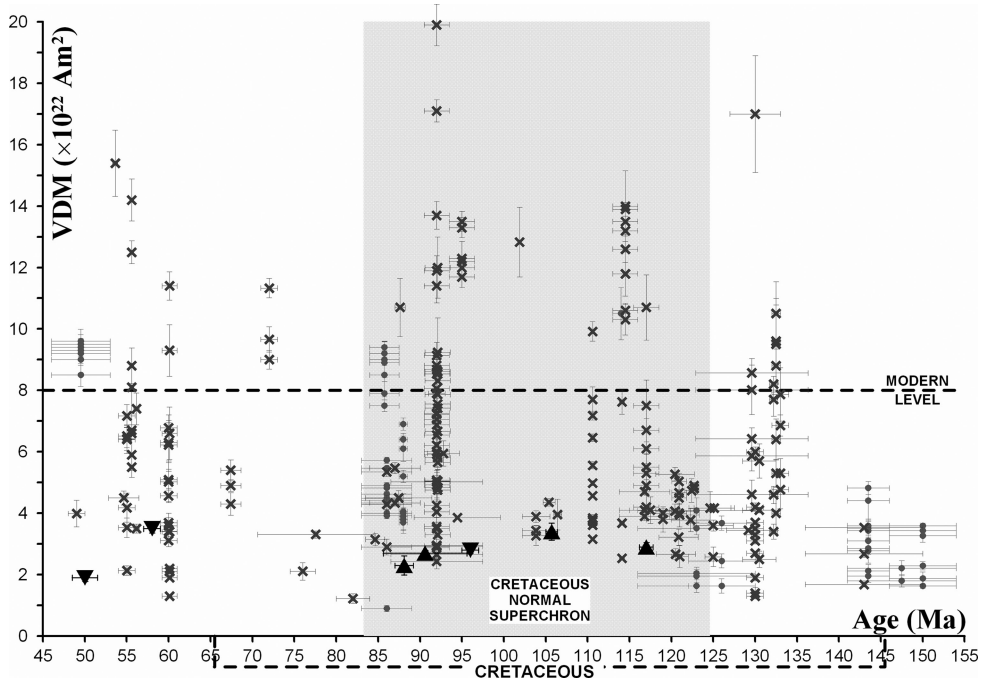


Figure 8. Summary of Palaeocene and Cretaceous VDMs from this study and from published sources obtained by the Thellier method and selected by the criteria (b) and (c). VDMs obtained in this study are represented by triangles, upward vertices of a triangle depict the normal polarity and vice versa. The data taken from Ade-Hall *et al.* (1968); Bol'shakov & Solodovnikov (1980, 1981, 1983); Solodovnikov (1998, 2001) passed the selection criteria (b) and they are shown with full circles. The data from Brandt *et al.* (2009); Carvallo *et al.* (2004); Cejudo Ruiz *et al.* (2006, 2009); Cottrell & Tarduno (2000); Goguitchaichvili *et al.* (2002b, 2004, 2008); Granot *et al.* (2007); Juarez *et al.* (1998); Kosterov *et al.* (1998); Qin *et al.* (2011); Riisager *et al.* (1999, 2001, 2004); Riisager & Abrahamsen (2000); Riisager *et al.* (2002); Selkin & Tauxe (2000); Shcherbakova *et al.* (2007, 2008, 2009, 2011); Sherwood *et al.* (1993); Shi *et al.* (2005); Smirnov & Tarduno (2003); Tanaka & Kono (2002); Tarduno *et al.* (2001; 2002); Tarduno & Cottrell (2005); Tauxe (2006); Tauxe & Staudigel (2004); Tsunakawa *et al.* (2009); Zhao *et al.* (2004); Zhu *et al.* (2001, 2003, 2004a,b,c, 2008) passed the more strict selection criteria (c) which includes the presence of successful pTRM check procedure and they are marked with crosses. The vertical error bars represent standard error of the mean SE $= \sigma(\mathrm{VDM})/\sqrt{n}$ where n is the number of samples used for the palaeointensity determination in the particular study. The horizontal bars display the uncertainty in the age determination.

Table 6. Mean values of VDM and their standard deviations calculated for data selected with the criteria (a), (b) and (c).

Time period	ν	N(a)	μ(a) ± SE	σ(a)	N(b)	μ(b) ± SE	σ(b)	N(c)	μ(c) ± SE	σ(c)
CNS	0.16	253	5.8 ± 0.2	3.1	168	6.2 ± 0.3	3.4	130	6.5 ± 0.3	3.5
Cretaceous excluding CNS	1.63	103	4.6 ± 0.3	2.5	69	4.6 ± 0.3	2.5	51	5.1 ± 0.4	2.7
Miocene	4.58	301	5.0 ± 0.2	3.3	123	4.6 ± 0.2	2.1	98	4.7 ± 0.2	2.3
154–172 Ma	9.78	62	3.0 ± 0.2	1.4	31	3.0 ± 0.3	1.6	-	-	-
Correlation with ν	Pearson p		−0.908	−0.755		−0.925	−0.915		−0.877	−0.928
			0.088	0.231		0.075	0.085		0.319	0.243
	Spearman p		−0.800	−0.400		−0.949	−1.000		−1.000	−1.000
			0.333	0.750		0.167	0.083		0.333	0.333
	Kendal p		−0.667	−0.333		−0.913	−1.000		−1.000	−1.000
			0.333	0.750		0.167	0.083		0.333	0.333

Note: ν is the number of reversals per Ma. N(a), N(b) and N(c) denote the number of VDM determinations for (a), (b) and (c) criteria sets are . μ(a), μ(b) and μ(c) are the mean values of VDMs, σ(a), σ(b) and σ(c) are the standard deviations calculated for (a), (b) and (c) criteria sets, respectively. SE is the standard error of mean, all values are given in 10^{22}Am^2 units. The low part of the table presents results of a correlation analysis using both linear (Pearson) or rank correlations (Kendal and Spearman) between reversal frequency and mean VDM and VDM scatter. For each statistics the figures in the first row present the values of the correlations, the figures in the second row gives the corresponding p-values obtained from this statistics.

determinations, Prévot *et al.* (1990) and Pick & Tauxe (1993) concluded that geomagnetic reversal rate and palaeointensity are decoupled. The comprehensive data compilation by Valet (2003) could not resolve the question whether a link between geomagnetic reversal rate and palaeointensity exists. Recent palaeointensity studies by Tarduno *et al.* (2006) and Tauxe (2006), however, favour the existence of an inverse relationship between reversal rate and geomagnetic field intensity.

An obvious way to deal with this problem is the comparison of palaeofield characteristics for different geological periods that have different rates of reversals. For the analysis of the statistical palaeofield characteristics we use both the mean VDM $= \mu$ and the standard deviation of VDMs $= \sigma$. The last parameter often is regarded as a measure of palaeosecular variation.

To search for the existence of links between reversal rate, average field strength and standard deviation of VDMs, we calculated first the average VDMs and their standard deviation throughout the Cretaceous (145.5–65.6 Ma), separately for the CNS (124–83 Myr) and for the rest of the Cretaceous where the rate of reversals is relatively low, about 1.7 per Ma. The same calculations were performed for the Miocene period (23–5.3 Ma), which had a high reversal rate of about 4.6 per Ma and for the Upper and Middle Jurassic where the reversal rate in the time interval (154–172) Ma was even higher, about 10 per Ma. The analysis was undertaken for the following three variants, with subsequently more strict criteria for the selection:

(a) All Thellier-type results are taken into account.

(b) The reliability criteria formulated by Perrin & Shcherbakov (1997) were used: only Thellier-type determinations are considered; at least three samples are used for the VDM calculation; the standard error of the mean VDM from a site should not exceed 15 per cent (the criterion of internal consistency). Underline that this way of selection uses not the standard deviation σ but an error of the estimate of the mean VDM, that is, $\sigma/\sqrt{n}$. This criterion has the advantage of automatic sample-sized dependent adjustment so a data set with the same standard deviation but smaller n has considerably less probability to pass the criteria. As an alternative way to account for the number of determinations, Paterson *et al.* (2010) proposed to use for the selection an sample-sized dependent adjustment to compare the maximum σ at the same 95 per cent confidence level for different data sets. It would be interesting to carry out a comparison of different sample-sized dependent dataset selection.

(c) The same as for (b), but the presence of the pTRM-check procedure is required.

The results of these calculations are shown in Table 6. Unfortunately, the Upper and Middle Jurassic are covered only with publications by Bol'shakov *et al.* (1987) and Sakai & Funaki (1988), which do not use the pTRM—check procedure. A few data for this period are obtained also by Tarduno & Cottrell (2005) but they pass only the criterion (a) as they have SE exceeding 15 per cent. Therefore, the case (c) is not presented in Table 6 for this period. Examination of the Table 6 suggests a tendency of decreasing of mean VDM with increasing reversal rate ν. Interestingly, this tendency becomes clearer when the more rigorous selection criterion (b) is applied. The case (c) shows weaker tendency, which is likely because of poor statistical representation. This observation agrees with the Cox (1968) hypothesis and is supported by the high enough correlation coefficients between ν and μ (Table 6).

Altogether with the correlations between ν and μ, one can see also the same behaviour of the standard deviation σ. It implies that the statistical parameters μ and σ are proportional to each other producing the correlation coefficients between μ and σ which are even higher than those of between ν and μ, namely Pearson's correlation coefficient between μ(b) and σ(b) is 0.96 while for the case (3) it amounts to 0.99. The hypothesis that the variability in palaeointensity can be proportional to the average field strength was suggested before by Constable *et al.* (1998) who concluded from their analysis of an 11-Myr-long Oligocene record from sediments that 'the average value of field intensity within each polarity interval is strongly correlated with its variability and weakly correlated with interval length'.

The comparison of the histograms of VDMs for the case (b) for the four periods (Fig. 9) clearly shows that the peak of VDM distribution does not shift substantially to bigger VDM values with the reversal rate increasing. Apparently, all correlations presented in Table 6, reflect mainly the relative increase in number of high values of VDMs which leads to enhancement of high VDM tail and asymmetry of distribution function of VDM. It is easy to see when comparing the distribution functions of VDMs presented in Fig. 9(e), moving from the Upper–Middle Jurassic to the CNS when reversals become less and less frequent.

However, we must admit that an application of statistical tests to justify the found here correlations between reversal rate and average VDM (or standard deviation) indicates that, although the

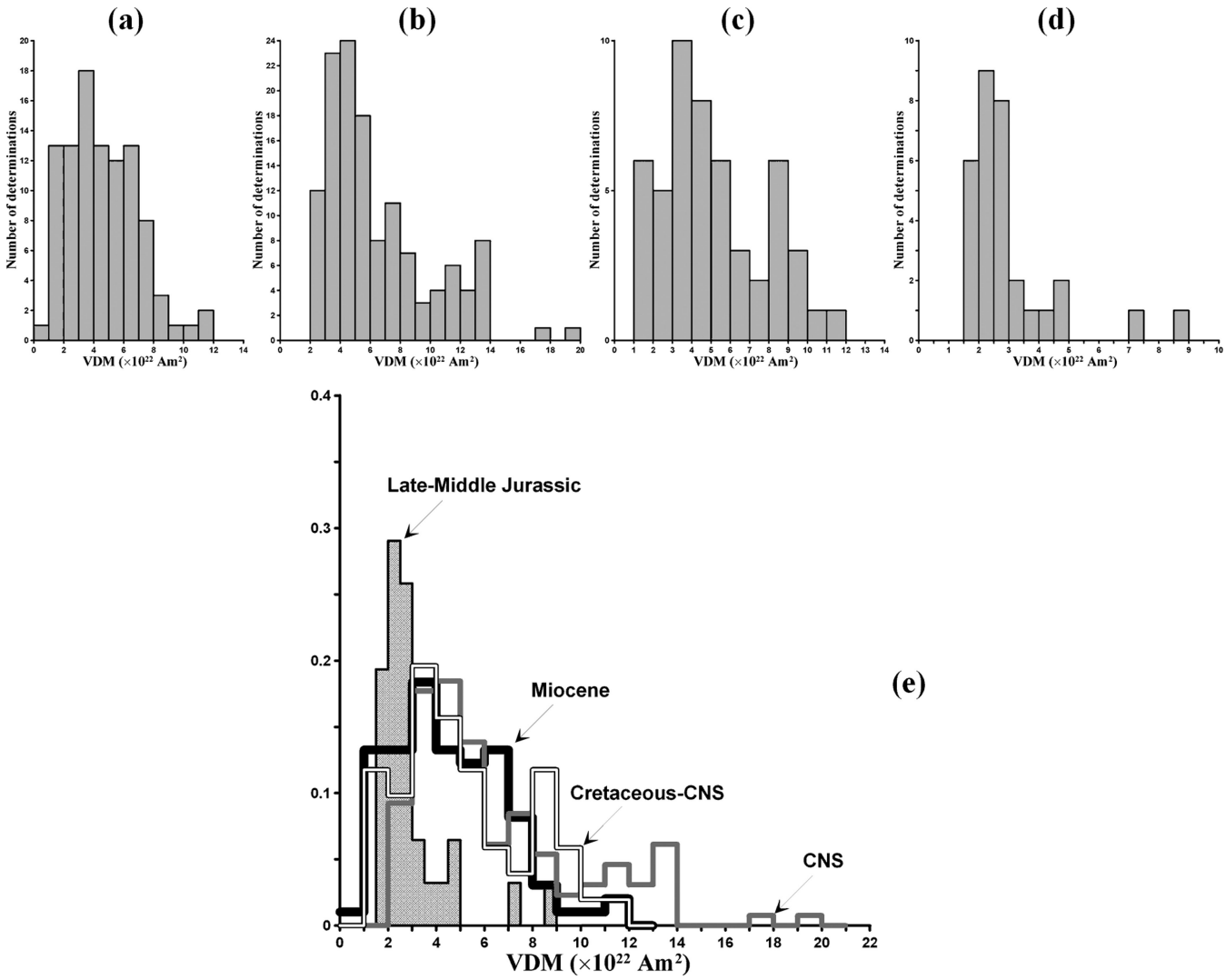


Figure 9. Top figures (a)–(d) are the histograms of palaeointensity data for the case (b) for different geological periods. VDM ($\times 10^{22}$ Am^2) is plotted on the horizontal axis, number of counts are plotted on the vertical axis. (a) Miocene (Calvo Rathert *et al.* 2009; Goguitchaichvili *et al.* 2000, 2002a, 2003; Juarez *et al.* 1998; Lawrence *et al.* 2009; Leonhardt *et al.* 2000, 2002; Leonhardt & Soffel 2006; Linder & Leonhardt 2009; Prévot *et al.* 1985; Riisager *et al.* 2000; Selkin & Tauxe 2000; Shcherbakova *et al.* 2010; Shi *et al.* 2002; Tauxe 2006; Tauxe *et al.* 2004), (b) Cretaceous Normal Superchron, (c) the Cretaceous excluding CNS, (d) Middle–Upper Jurassic (Bol'shakov *et al.* 1987; Sakai & Funaki 1988). The references for the Cretaceous are given in the caption for the Fig. 8. Fig. 9(e) in the bottom presents the distribution functions of VDMs for these periods obtained by normalizing the histograms 9(a)–(d).

correlations are strong, they are not significant at the 5 per cent significance level. This is demonstrated by both linear (Pearson) or rank correlations (Kendal and Spearman) analyses (Table 6). The same is true for the correlation between μ and σ for the case (b) though for the case (c) the significance is higher than the 5 per cent level. This negative outcome is not a surprise as the data sets under the consideration are too small for convincing statistics even if the result of a test is positive on the 5 per cent significance level. Besides, any statistical tests of VDM data presented in Table 6, can in general provide only a rough justification to a statistical hypothesis as the databases at hand are still far from the completeness and their statistical properties can hardly be described precisely anyway. In particular, turn attention to the fact that the datasets over every geological period used here for the calculations have irregular histograms and certainly cannot be described by the normal distribution (Fig. 9). Note that non-symmetrical non-Gaussian like those seen in Fig. 9, naturally suggests the existence of (μ, σ) correlation. This property was already exploited by Constable *et al.* (1998) who proposed the gamma distribution

$$p(t) = \frac{\lambda^k}{\Gamma(k)} t^{k-1} \exp(-\lambda t)$$

to describe formally the observed VDM distribution for sediment record. Here k is a shape parameter and λ is a scale parameter. It is easy to see that for this distribution $\mu = \sigma/\sqrt{k}$, that is, the mean value μ and the variance σ are really directly proportional to each other.

As the final remark, note that there is a possibility that the asymmetry and thus the higher value for the CNS can be simply a result of poor temporal sampling but not a real feature of VDM values distribution for the CNS. As is evident from the analysis given above, the state of art in palaeomagnetism does not allow to argue conclusively that the average VDM value indeed represents the time-averaged VDM for the CNS (kind of ergodic hypothesis). Rather, we must admit that the question of what are VDM

distributions at different geological periods is now one of main challenges to palaeomagnetism.

5 CONCLUSIONS

Results of palaeomagnetic analysis presented in this study agree well with the previous palaeodirection determinations for the Western Antarctic. Along with normal polarity, one Cretaceous site (Petermann Island) allocated a reverse polarity recorded in high-temperature component of NRM. An unambiguous interpretation of this observation requires further investigation.

Seven reliable palaeointensity determinations for seven sites from this area are obtained using in total 139 sister cubes of 75 different samples. The resulting VDM values range from 1.9 to 3.5 (10^{22} Am^2). Analysis of palaeointensity data available from the updated to the present day IAGA World palaeointensity database for four geological periods (CNS, the rest of Cretaceous, Miocene and Late-Middle Jurassic) with different frequency of reversals proved the existence of strong inverse correlations between the mean VDM versus the rate of inversions. The same is found for standard deviation of VDMs. However, due to shortage of data, the correlations are not significant at the 5 per cent significance level, hence, palaeointensity data cannot yet firmly confirm the existence of a link between the mean VDM (or standard deviation) versus the rate of reversals.

ACKNOWLEDGMENTS

The authors are very grateful to Greig Paterson for the valuable comments and suggestions made to this paper. The work was supported by Russian Fund of Basic Research, grant no. 09-05-00878 and Ukrainian Fund of Basic Research, grant F no. Φ40.6/030.

REFERENCES

Ade-Hall, J.M., Khan, M.A., Dagley, P. & Wilson, R.L., 1968. A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow from Skye, Scotland, *Geophys. J. R. astron. Soc.,* **16**(4), 375–415, doi:10.1111/j.1365-246X.1968.tb00231.x.

Besse, J. & Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, *J. geophys. Res.,* **107**(B11), 2300, doi:10.1029/2000JB000050.

Biggin, A.J. & Thomas, D.N., 2003. The application of acceptance criteria to results of Thellier palaeointensity experiments performed on samples with pseudo-single-domain-like characteristics, *Phys. Earth planet. Inter.,* **138**(3–4), 279–287, doi:10.1016/s0031-9201(03)00127-4.

Biggin, A.J., McCormack, A. & Roberts, A., 2010. Paleointensity database updated and upgraded, *EOS, Trans. Am. geophys. Un.,* **91**(2), 5.

Blundell, D.J., 1962. Palaeomagnetic investigations in the Falkland Islands Dependencies, *Br. Antarct. Surv. Sci. Rep.,* **39,** 1–24.

Bol'shakov, A.S. & Solodovnikov, G.M., 1980. Geomagnetic field strength in the Late Jurassic, *Izv. Phys. Solid Earth,* **16,** 848–857.

Bol'shakov, A.S. & Solodovnikov, G.M., 1981. Intensity of the geomagnetic field in the last 400 million years, *Dokl. Earth Sci.,* **260,** 24–27.

Bol'shakov, A.S. & Solodovnikov, G.M., 1983. Geomagnetic field intensity in Armenia in the Late Jurassic and Early Cretaceous, *Izv. Phys. Solid Earth,* **19,** 976–982.

Bol'shakov, A.S., Solodovnikov, G.M. & Vinogradov, Y.K., 1987. Paleointensity of the geomagnetic field in the Early and Middle Jurassic, *Izv. Phys. Solid Earth,* **23,** 324–333.

Brandt, D., Hartmann, G.A., Yokoyama, E., Catelani, E.L. & Trindade, R.I.F., 2009. Paleointensity data from Early Cretaceous Ponta Grossa dikes (Brazil) using a multisample method, *EPS,* **61**(1), 41–49.

Calvo Rathert, M., Goguitchaichvili, A. & Vegas-Tubía, N., 2009. A paleointensity study on middle Miocene to Pliocene volcanic rocks from south-eastern Spain, *Earth, Planets and Space,* **61**(1), 61–69.

Carvallo, C., Özdemir, Ö. & Dunlop, D.J., 2004. Palaeointensity determinations, palaeodirections and magnetic properties of basalts from the Emperor seamounts, *Geophys. J. Int.,* **156**(1), 29–38, doi:10.1111/j.1365-246X.2004.02110.x.

Carvallo, C., Roberts, A.P., Leonhardt, R., Laj, C., Kissel, C., Perrin, M. & Camps, P., 2006. Increasing the efficiency of paleointensity analyses by selection of samples using first-order reversal curve diagrams, *J. geophys. Res.,* **111**(B12), B12103, doi:10.1029/2005jb004126.

Cejudo Ruiz, R., Goguitchaichvili, A., Geuna, S.E., Alva Valdivia, L.M., Solé, J. & Morales Contreras, J., 2006. Early cretaceous absolute geomagnetic paleointensities from Córdoba Province (Argentina), *Earth, Planet and Space,* **58**(10), 1333–1339.

Cejudo Ruiz, R., Goguitchaichvili, A., Morales Contreras, J., Trindade, R.I.F., Alva Valdivia, L.M. & Urrutia Fucugauchi, J., 2009. Absolute Thellier paleointensities from Ponta Grossa dikes (southern Brazil) and the early Cretaceous geomagnetic field strength, *Geofis. Intl.,* **48**(2), 243–251.

Chadima, M. & Hrouda, F., 2006. Remasoft 3.0 a user-friendly paleomagnetic data browser and analyzer, *Travaux Géophysiques,* **XXVII,** 20–21.

Coe, R.S., 1967. The determination of paleointensities of the Earth's magnetic field with special emphasize on mechanisms which could cause nonideal behavior in Thellier's method, *J. Geomagn. Geoelec.,* **19**(3), 157–179.

Coe, R.S., Grommé, S. & Mankinen, E.A., 1978. Geomagnetic paleointensities from radiocarbon-dated lava flows on Hawaii and the question of the Pacific nondipole low, *J. geophys. Res.,* **83**(B4), 1740–1756, doi:10.1029/JB083iB04p01740.

Constable, C.G., Tauxe, L. & Parker, R.L., 1998. Analysis of 11 Myr of geomagnetic intensity variation, *J. geophys. Res.,* **103**(B8), 17735–17748, doi:10.1029/98jb01519.

Cottrell, R.D. & Tarduno, J.A., 2000. In search of high-fidelity geomagnetic paleointensities: a comparison of single plagioclase crystal and whole rock Thellier-Thellier analyses, *J. geophys. Res.,* **105**(B10), 23579–23594, doi:10.1029/2000jb900219.

Cox, A., 1968. Lengths of geomagnetic polarity intervals, *J. geophys. Res.,* **73**(10), 3247–3260, doi:10.1029/JB073i010p03247.

Curtis, R., 1966. The petrology of the Graham Coast, Graham Land, *Br. Antarct. Surv. Sci. Rep.,* **50,** 1–51.

Dalziel, I.W.D. & Elliot, D.H., 1982. West Antarctica: problem child of Gondwanaland, *Tectonics,* **1**(1), 3–19, doi:10.1029/TC001i001p00003.

Dalziel, I.W.D., Lowtie, W., Kligfield, R. & Opdyke, N.D., 1973. Paleomagnetic data from the southernmost Andes and the Antarctandes, in *Implications of Continental Drift to the Earth Sciences,* pp. 87–101, eds. Tarling, D.H. & Runcorn, S.K. Academic Press, New York, NY.

Day, R., Fuller, M. & Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence, *Phys. Earth planet. Inter.,* **13**(4), 260–267, doi:10.1016/0031-9201(77)90108-x.

Dodson, M.H. & McClelland-Brown, E., 1980. Magnetic blocking temperatures of single-domain grains during slow cooling, *J. geophys. Res.,* **85**(B5), 2625–2637, doi:10.1029/JB085iB05p02625.

Ferraccioli, F., Jones, P.C., Vaughan, A.P. & Leat, P.T., 2006. New aerogeophysical view of the Antarctic Peninsula: more pieces, less puzzle, *Geophys. Res. Lett.,* **33,** doi:10.1029/2005GL024636.

Fleming, E.A. & Thomson, J.W., 1979. Northern Graham Land and South Shetland Islands, *in British Antarctic Territory Geological Map*, 1:500,000 scale, British Antarctic Survey, Cambridge.

Gapeev, A.K. & Tselmovich, V.A., 1983. Microtexture of synthetic titanomagnetite oxidized at high partial pressures of oxygen, *Izv. Phys. Solid Earth,* **19**(22), 983–986.

Gapeev, A.K. & Tselmovich, V.A., 1986. Microstructure of natural heterophasically oxidized titanomagnetites, *Izv. Phys. Solid Earth,* **22**(4), 331–334.

Gapeev, A.K. & Tselmovich, V.A., 1992. The microstructure and domain structure of multiphase oxidized titanomagnetites, *Phys. Earth planet. Inter.,* **70**(3–4), 243–247, doi:10.1016/0031-9201(92)90190-7.

Gladkochub, D.P., Bakhmutov, V.G., Vodovozov, V.Y. & Vaschenko, V., 2009. West Antarctica Andean complex: age, compound, geodynamic position, *in Geology of the Earth's Polar Regions, XLII Tectonic Meeting*, *Moskow*, *Russia*, Vol. 1, pp. 132–134, ed. Karjakin, Y., GEOS, Moscow (in Russian).

Goguitchaichvili, A., Alva Valdivia, L.M., Urrutia Fucugauchi, J., Morales Contreras, J. & Ferrari, L., 2000. Absolute palaeointensity results from the Trans-Mexican Volcanic Belt: implications for the late Miocene geomagnetic field strength, *Geophys. J. Int.,* **143**(3), 977–984, doi:10.1046/j.0956-540X.2000.01301.x.

Goguitchaichvili, A., Alva Valdivia, L.M., Rosas Elguera, J., Urrutia Fucugauchi, J., Angel Cervantes, M. & Morales Contreras, J., 2002a. Paleosecular variation record of geomagnetic full vector during late Miocene, from the Nayarit area, Mexico, *Phys. Earth planet. Inter.* **134**(1–2), 71–88, doi:10.1016/s0031-9201(02)00096-1.

Goguitchaichvili, A., Alva Valdivia, L.M., Urrutia Fucugauchi, J., Morales Contreras, J. & Ferreira Lopes, O., 2002b. On the reliability of Mesozoic Dipole Low: new absolute paleointensity results from Paraná Flood Basalts (Brazil), *Geophys. Res. Lett.,* **29**(13), 1655, doi:10.1029/2002gl015242.

Goguitchaichvili, A., Morales Contreras, J., Cañon-Tapia, E. & Negrete, R., 2003. Geomagnetic field strength during late Miocene: first paleointensity results from Baja California, *J. geophys. Res.,* **108**(B2), 2114, doi:10.1029/2002jb002081.

Goguitchaichvili, A., Alva Valdivia, L.M., Rosas Elguera, J., Urrutia Fucugauchi, J. & Solé, J., 2004. Absolute geomagnetic paleointensity after the Cretaceous Normal Superchron and just prior to the Cretaceous-Tertiary transition, *J. geophys. Res.,* **109**(B1), B01105, doi:10.1029/2003jb002477.

Goguitchaichvili, A., Cejudo Ruiz, R., Sanchez Bettucci, L., Aguilar Reyes, B., Alva Valdivia, L.M., Urrutia Fucugauchi, J., Morales Contreras, J. & Calvo Rathert, M., 2008. New absolute paleointensity results from the Parana Magmatic Province (Uruguay) and the Early Cretaceous geomagnetic paleofield, *Geochem. Geophys. Geosyst.,* **9**(11), Q11008, doi:10.1029/2008gc002102.

Gradstein, F.M., Ogg, J.G. & Smith, A.G., 2004. *A Geologic Time Scale 2004,* Cambridge University Press, Cambridge, 589pp.

Granot, R., Tauxe, L., Gee, J.S. & Ron, H., 2007. A view into the Cretaceous geomagnetic field from analysis of gabbros and submarine glasses, *Earth planet. Sci. Lett.,* **256**(1–2), 1–11, doi:10.1016/j.epsl.2006.12.028.

Grikurov, G.E., 1973. *Geology of the Antarctic Peninsula,* pp. 120, Nauka, Moscow (in Russian).

Grunow, A., 1993. New paleomagnetic data from Antarctic Peninsula and their tectonic implication, *J. geophys. Res.,* **98**(B8), 13815–13833, doi:10.1029/93JB01089.

Guenthner, W.R., Barbeau, D.L., Jr, Reiners, P.W. & Thomson, S.N., 2010. Slab window migration and terrane accretion preserved by low-temperature thermochronology of a magmatic arc, northern Antarctic Peninsula, *Geochem. Geophys. Geosyst.,* **11**(3), Q03001, doi:10.1029/2009gc002765.

Halgedahl, S.L., Day, R. & Fuller, M., 1980. The effect of cooling rate on the intensity of weak-field TRM in single-domain magnetite, *J. geophys. Res.,* **85**(B7), 3690–3698, doi:10.1029/JB085iB07p03690.

Heunemann, C., Krása, D., Soffel, H.C., Gurevitch, E. & Bachtadse, V., 2004. Directions and intensities of the Earth's magnetic field during a reversal: results from the Permo-Triassic Siberian trap basalts, Russia, *Earth planet. Sci. Lett.,* **218**(1–2), 197–213, doi:10.1016/s0012-821x(03)00642-3.

Juarez, M.T., Tauxe, L., Gee, J.S. & Pick, T., 1998. The intensity of the Earth's magnetic field over the past 160 million years, *Nature,* **394**(6696), 878–881, doi:10.1038/29746.

Kellogg, K., 1980. Paleomagnetic evidence for oroclinal bending of the southern Antarctic Peninsula, *Geol. Soc. Am. Bull.,* **91,** 414–420, doi:10.1130/0016-7606(1980)91<414:PEFOBO>2.0.CO;2.

Kellogg, K. & Reynolds, R., 1978. Paleomagnetic results from the Lassiter Coast, Antarctica, and a test for oroclinal bending of the Antarctic Peninsula, *J. geophys. Res.,* **83**(B5), 2293–2299, doi:10.1029/JB083iB05p02293.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astron. Soc.,* **62,** 699–718, doi:10.1111/j.1365-246X.1980.tb02601.x.

Kosterov, A.A., Perrin, M., Glen, J.M. & Coe, R.S., 1998. Paleointensity of the Earth's magnetic field in Early Cretaceous time: the Paraná Basalt, Brazil, *J. geophys. Res.,* **103**(B5), 9739–9753, doi:10.1029/98jb00022.

Lawrence, K.P., Tauxe, L., Staudigel, H., Constable, C.G., Koppers, A., McIntosh, W. & Johnson, C.L., 2009. Paleomagnetic field properties at high southern latitude, *Geochem. Geophys. Geosyst.,* **10**(1), Q01005, doi:10.1029/2008gc002072.

Leat, P.T., Scarrow, J.H. & Millar, L.L., 1995. On the Antarctic Peninsula batholith, *Geol. Mag.,* **132**(4), 399–412, doi:10.1017/S0016756800021464.

Leonhardt, R. & Soffel, H.C., 2006. The growth, collapse and quiescence of Teno volcano, Tenerife: new constraints from paleomagnetic data, *Int. J. Earth Sci.,* **95**(6), 1053–1064, doi:10.1007/s00531-006-0089-3.

Leonhardt, R., Hufenbecher, F., Heider, F. & Soffel, H.C., 2000. High absolute paleointensity during a mid Miocene excursion of the Earth's magnetic field, *Earth planet. Sci. Lett.,* **184**(1), 141–154, doi:10.1016/s0012-821x(00)00311-3.

Leonhardt, R., Matzka, J., Hufenbecher, F., Soffel, H.C. & Heider, F., 2002. A reversal of the Earth's magnetic field recorded in mid-Miocene lava flows of Gran Canaria: paleodirections, *J. geophys. Res.,* **107**(B1), 2024, doi:10.1029/2001jb000322.

Linder, J. & Leonhardt, R., 2009. Paleomagnetic full vector record of four consecutive Mid Miocene geomagnetic reversals, *Phys. Earth planet. Inter.,* **177**(1–2), 88–101, doi:10.1016/j.pepi.2009.07.013.

Longshaw, S.K. & Griffiths, D.H., 1983. A palaeomagnetic study of Jurassic rocks from the Antarctic Peninsula and its implications, *J. Geol. Soc.,* **140**(6), 945–954, doi:10.1144/gsjgs.140.6.0945.

McClelland-Brown, E., 1984. Experiments on TRM intensity dependence on cooling rate, *Geophys. Res. Lett.,* **11**(3), 205–208, doi:10.1029/GL011i003p00205.

Muxworthy, A.R., 2010. Revisiting a domain-state independent method of palaeointensity determination, *Phys. Earth planet. Inter.,* **179**(1–2), 21–31, doi:10.1016/j.pepi.2010.01.003.

Nawrocki, J., Panczyk, M. & Williams, I.S., 2010. Isotopic ages and palaeomagnetism of selected magmatic rocks from King George Island (Antarctic Peninsula), *J. Geol. Soc.,* **167**(5), 1063–1079, doi:10.1144/0016-76492009-177.

Néel, L., 1955. Some theoretical aspects of rock-magnetism, *Adv. Phys.,* **4**(14), 191–243, doi:10.1080/00018735500101204.

Pankhurst, R.J., 1982. Rb-Sr geochronology of Graham Land, Antarctica, *J. Geol. Soc.,* **139**(6), 701–711, doi:10.1144/gsjgs.139.6.0701.

Pankhurst, R.J., 1983. Rb-Sr constraints on the ages of basement rocks of the Antarctic Peninsula, *in Antarctic Earth Science: Fourth International Symposium,* Adelaide, Australia, 1983, pp. 367–371, eds. Oliver, R.L., James, P.R. & Jago, J.B. Cambridge University Press, New York.

Pares, J.M. & Dinares, J.T., 1999. Datos paleomagnéticos del sustrato rocoso de la isla de Livinsgton (Península Antártica): implicaciones tectónicas en la evolución neógena, *Acta Geol. Hisp.,* **34**(4), 339–352 (in Spanish).

Paterson, G.A., 2011. A simple test for the presence of multidomain behavior during paleointensity experiments, *J. geophys. Res.,* **116**(B10), B10104, doi:10.1029/2011jb008369.

Paterson, G.A., Heslop, D. & Muxworthy, A.R., 2010. Deriving confidence in paleointensity estimates, *Geochem. Geophys. Geosyst.,* **11**(7), Q07Z18, doi:10.1029/2010gc003071.

Perrin, M. & Schnepp, E., 2004. IAGA paleointensity database: distribution and quality of the data set, *Phys. Earth planet. Inter.,* **147**(2–3), 255–267, doi:10.1016/j.pepi.2004.06.005.

Perrin, M. & Shcherbakov, V.P., 1997. Paleointensity of the Earth's magnetic field for the past 400 Ma: evidence for a Dipole structure during the Mesozoic Low, *J. Geomagn. Geoelec.,* **49**(4), 601–614.

Pick, T. & Tauxe, L., 1993. Geomagnetic palaeointensities during the Cretaceous normal superchron measured using submarine basaltic glass, *Nature,* **366**(6452), 238–242, doi:10.1038/366238a0.

Poblete, F., Arriagada, C., Roperch, P., Astudillo, N., Herve, F., Kraus, S. & Le Roux, J.P., 2011. Paleomagnetism and tectonics of the South Shetland

Islands and the northern Antarctic Peninsula, *Earth planet. Sci. Lett.,* **302**(3–4), 299–313, doi:10.1016/j.epsl.2010.12.019.

Prévot, M., Mankinen, E.A., Coe, R.S. & Grommé, S.C., 1985. The Steens Mountain (Oregon) Geomagnetic Polarity Transition 2. Field Intensity Variations and Discussion of Reversal Models, *J. geophys. Res.,* **90**(B12), 10417–10448.

Prévot, M., Derder, M.E.-M., McWilliams, M. & Thompson, J., 1990. Intensity of the Earth's magnetic field: evidence for a Mesozoic dipole low, *Earth planet. Sci. Lett.,* **97**(1–2), 129–139, doi:10.1016/0012-821x(90)90104-6.

Qin, H., He, H., Liu, Q. & Cai, S., 2011. Palaeointensity just at the onset of the Cretaceous normal superchron, *Phys. Earth planet. Inter.,* **187**(3–4), 199–211, doi:10.1016/j.pepi.2011.05.009.

Rex, D.C., 1976. Geochronology in relation to the stratigraphy of the Antarctic Peninsula, *Br. Antarct. Surv. Bull.,* **43,** 49–58.

Riisager, P. & Abrahamsen, N., 2000. Palaeointensity of West Greenland Palaeocene basalts: asymmetric intensity around the C27n-C26r transition, *Phys. Earth planet. Inter.,* **118**(1–2), 53–64, doi:10.1016/s0031-9201(99)00125-9.

Riisager, J., Riisager, P. & Perrin, M., 1999. Palaeodirectional and palaeointensity results of Paleocene and Eocene basalts from West Greenland, *Bull. Geol. Soc. Den.,* **46,** 69–78.

Riisager, J., Perrin, M., Riisager, P. & Ruffet, G., 2000. Paleomagnetism, paleointensity and geochronology of Miocene basalts and baked sediments from Velay Oriental, French Massif Central, *J. geophys. Res.,* **105**(B1), 883–896, doi:10.1029/1999jb900337.

Riisager, J., Perrin, M., Riisager, P. & Vandamme, D., 2001. Palaeomagnetic results and palaeointensity of Late Cretaceous Madagascan basalt, *J. Afr. Earth. Sci.,* **32**(3), 503–518, doi:10.1016/s0899-5362(01)90111-3.

Riisager, P., Riisager, J., Abrahamsen, N. & Waagstein, R., 2002. Thellier palaeointensity experiments on Faroes flood basalts: technical aspects and geomagnetic implications, *Phys. Earth planet. Inter.,* **131**(2), 91–100, doi:10.1016/s0031-9201(02)00031-6.

Riisager, J., Riisager, P., Zhao, X., Coe, R.S. & Pedersen, A.K., 2004. Paleointensity during a chron C26r excursion recorded in west Greenland lava flows, *J. geophys. Res.,* **109**(B4), B04107, doi:10.1029/2003jb002887.

Riley, T.R., Leat, P.T., Pankhurst, R.J. & Harris, C., 2001. Origins of Large Volume Rhyolitic Volcanism in the Antarctic Peninsula and Patagonia by Crustal Melting, *J. Petrol.,* **42**(6), 1043–1065, doi:10.1093/petrology/42.6.1043.

Sakai, H. & Funaki, M., 1988. Paleomagnetic study of the Beacon supergroup in Antartica: remagnetization in the Jurassic time, *Proc. NIPR Symp. Antarct. Geosci.,* **2,** 46–54.

Selkin, P.A. & Tauxe, L., 2000. Long-term variations in palaeointensity, *Philos. Trans. R. Soc. London, Ser. A,* **358**(1768), 1065–1088, doi:10.1098/rsta.2000.0574.

Shcherbakov, V.P. & Shcherbakova, V.V., 2001. On the suitability of the Thellier method of palaeointensity determinations on pseudo-single-domain and multidomain grains, *Geophys. J. Int.,* **146**(1), 20–30, doi:10.1046/j.0956-540x.2001.01421.x.

Shcherbakova, V.V., Shcherbakov, V.P. & Heider, F., 2000. Properties of partial thermoremanent magnetization in pseudosingle domain and multidomain magnetite grains, *J. geophys. Res.,* **105**(B1), 767–781, doi:10.1029/1999jb900235.

Shcherbakova, V.V., Asanidze, B.Z., Shcherbakov, V.P. & Zhidkov, G.V., 2007. Geomagnetic field paleointensity in the cretaceous from Upper Cretaceous rocks of Georgia, *Izv. Phys. Solid Earth,* **43**(11), 951–959, doi:10.1134/s1069351307110055.

Shcherbakova, V.V., Zhidkov, G.V. & Shcherbakov, V.P., 2008. Revised determination of the paleointensity in the Cretaceous from the collection of A.S. Bol'shakov and G.M. Solodovnikov, *Izv. Phys. Solid Earth,* **44**(10), 816–821, doi:10.1134/s1069351308100091.

Shcherbakova, V.V., Perrin, M., Shcherbakov, V.P., Pavlov, V.E., Ayvaz'yan, A. & Zhidkov, G.V., 2009. Rock magnetic and paleointensity results from Mesozoic baked contacts of Armenia, *Earth, Planet and Space,* **61**(1), 23–39.

Shcherbakova, V.V., Shcherbakov, V.P., Bretshtein, Y.S. & Zhidkov, G.V., 2010. Paleointensity and paleodirection of the geomagnetic field in the middle Miocene: evidence from late cenozoic volcanites of primorye, *Izv. Phys. Solid Earth,* **46**(12), 1035–1051, doi:10.1134/s1069351310120025.

Shcherbakova, V.V., Kovalenko, D.V., Shcherbakov, V.P. & Zhidkov, G.V., 2011. Paleointensity of the geomagnetic field in the Cretaceous (from Cretaceous rocks of Mongolia), *Izv. Phys. Solid Earth,* **47**(9), 775–791.

Sherwood, G., Shaw, J., Baer, G. & Mallik, S.B., 1993. The strength of the geomagnetic field during the Cretaceous quiet zone: palaeointensity results from Israeli and Indian lavas, *J. Geomagn. Geoelec.,* **45**(4), 339–360.

Shi, R., Zhu, R., Pan, Y., Shi, G. & Hoffman, K.A., 2002. Paleointensity study of Early Miocene lavas from Pingzhuang, Inner Mongolia, China, *Geophys. Res. Lett.,* **29**(21), 2026, doi:10.1029/2002gl015990.

Shi, R., Hill, M.J., Zhu, R., He, H. & Shaw, J., 2005. 40Ar/39Ar dating and preliminary paleointensity determination on a single lava flow from Chifeng, Inner Mongolia, *Phys. Earth planet. Inter.,* **152**(1–2), 78–89, doi:10.1016/j.pepi.2005.06.005.

Smirnov, A.V. & Tarduno, J.A., 2003. Magnetic hysteresis monitoring of Cretaceous submarine basaltic glass during Thellier paleointensity experiments: evidence for alteration and attendant low field bias, *Earth planet. Sci. Lett.,* **206**(3–4), 571–585, doi:10.1016/s0012-821x(02)01123-8.

Smirnov, A.V. & Tarduno, J.A., 2005. Thermochemical remanent magnetization in Precambrian rocks: are we sure the geomagnetic field was weak?, *J. geophys. Res.,* **110**(B6), B06103, doi:10.1029/2004jb003445.

Solodovnikov, G.M., 1998. The intensity of the Eocene geomagnetic field, *Izv. Phys. Solid Earth,* **34**(10), 865–869.

Solodovnikov, G.M., 2001. Determination of the geomagnetic field intensity in the Santonian-Coniacian (Upper Cretaceous) from an effusive section in Azerbaijan, *Izv. Phys. Solid Earth,* **37**(7), 600–605.

Tanaka, H. & Kono, M., 2002. Paleointensities from a Cretaceous basalt platform in Inner Mongolia, northeastern China, *Phys. Earth planet. Inter.,* **133**(1–4), 147–157, doi:10.1016/s0031-9201(02)00091-2.

Tangeman, J.A., Mukasa, S.B. & Grunow, A.M., 1996. Zircon U-Pb geochronology of plutonic rocks from the Antarctic Peninsula: confirmation of the presence of unexposed Paleozoic crust, *Tectonics,* **15**(6), 1309–1324, doi:10.1029/96tc00840.

Tarduno, J.A. & Cottrell, R.D., 2005. Dipole strength and variation of the time-averaged reversing and nonreversing geodynamo based on Thellier analyses of single plagioclase crystals, *J. geophys. Res.,* **110**(B11), B11101, doi:10.1029/2005jb003970.

Tarduno, J.A., Cottrell, R.D. & Smirnov, A.V., 2001. High geomagnetic intensity during the mid-Cretaceous from Thellier analyses of single plagioclase crystals, *Science,* **291**(5509), 1779–1783, doi:10.1126/science.1057519.

Tarduno, J.A., Cottrell, R.D. & Smirnov, A.V., 2002. The Cretaceous superchron geodynamo: Observations near the tangent cylinder, *PNAS,* **99**(22), 14020–14025, doi:10.1073/pnas.222373499.

Tarduno, J.A., Cottrell, R.D. & Smirnov, A.V., 2006. The paleomagnetism of single silicate crystals: recording geomagnetic field strength during mixed polarity intervals, superchrons, and inner core growth, *Rev. Geophys.,* **44**(1), RG1002, doi:10.1029/2005rg000189.

Tauxe, L., 2006. Long-term trends in paleointensity: the contribution of DSDP/ODP submarine basaltic glass collections, *Phys. Earth planet. Inter.,* **156**(3–4), 223–241, doi:10.1016/j.pepi.2005.03.022.

Tauxe, L. & Staudigel, H., 2004. Strength of the geomagnetic field in the Cretaceous Normal Superchron: new data from submarine basaltic glass of the Troodos Ophiolite, *Geochem. Geophys. Geosyst.,* **5**(2), Q02H06, doi:10.1029/2003gc000635.

Tauxe, L., Gans, P. & Mankinen, E.A., 2004. Paleomagnetism and 40Ar/39Ar ages from volcanics extruded during the Matuyama and Brunhes Chrons near McMurdo Sound, Antarctica, *Geochem. Geophys. Geosyst.,* **5**(6), Q06H12, doi:10.1029/2003gc000656.

Thomson, M.R.A. & Pankhurst, R.J., 1983. Age of post-Gondwanian calc-alkaline volcanism in the Antarctic Peninsula region, *in Antarctic Earth Science: Fourth International Symposium*, *Adelaide*, *Australia*, pp. 328–333, eds. Oliver, R.L., James, P.R. & Jago, J.B. Cambridge University Press, New York, NY.

Torsvik, T.H., Gaina, C. & Redfield, T.F., 2008. Antarctica and global paleogeography: from Rodinia, through Gondwanaland and Pangea, to the

birth of the Southern Ocean and the opening of gateways, Antarctica: a keystone in a changing world, in *Proceedings of the 10th International Symposium on Antarctic Earth Sciences,* Santa Barbara, CA, pp. 125–140, eds. Cooper, A.K., Barret, P.J., Stagg, H., Storey, B., Stump, E., Wise, W. & the 10th ISAES Editorial Team. National Academies Press, Washington, DC.

Tsunakawa, H., Wakabayashi, K.-i., Mochizuki, N., Yamamoto, Y., Ishizaka, K., Hirata, T., Takahashi, F. & Seita, K., 2009. Paleointensity study of the middle Cretaceous Iritono granite in northeast Japan: implication for high field intensity of the Cretaceous normal superchron, *Phys. Earth planet. Inter.,* **176**(3–4), 235–242, doi:10.1016/j.pepi.2009.07.001.

Valencio, D.A., Mendia, J.E. & Vilas, J.F., 1979. Palaeomagnetism and K–Ar age of Mesozoic and Cenozoic igneous rocks from Antarctica, *Earth planet. Sci. Lett.,* **45**(1), 61–68, doi:10.1016/0012-821x(79)90107-9.

Valet, J.-P., 2003. Time variations in geomagnetic intensity, *Rev. Geophys.,* **41**(1), 1004, doi:10.1029/2001rg000104.

Vaughan, A.P.M. & Storey, B.C., 2000. The Eastern Palmer Land shear zone: a new terrane accretion model for the Mesozoic development of the Antarctic Peninsula, *J. Geol. Soc.,* **157**(6), 1243–1256, doi:10.1144/jgs.157.6.1243.

Watts, D.R., Watts, G.C. & Bramall, A.M., 1984. Cretaceous and Early Tertiary paleomagnetic results from the Antarctic Peninsula, *Tectonics,* **3**(3), 333–346, doi:10.1029/TC003i003p00333.

Weaver, S.D., Saunders, A.D. & Tarney, J., 1982. Mesozoic-Cenozoic volcanism in the South Shetland Islands and the Antarctic Peninsula: geochemical nature and plate tectonic significance, in *Antarctic Geoscience: Symposium on Antarctic Geology and Geophysics*, Madison, WI, pp. 263–273, ed. Craddock, C., University of Wisconsin Press, Madison, WI.

Willan, R.C.R. & Kelley, S.P., 1999. Mafic dike swarms in the South Shetland Islands volcanic arc: unravelling multiepisodic magmatism related to subduction and continental rifting, *J. geophys. Res.,* **104**(B10), 23051–23068, doi:10.1029/1999jb900180.

Wilson, R.L., 1961. The thermal demagnetization of natural magnetic moments in rocks, *Geophys. J. R. astron. Soc.,* **5**(1), 45–58, doi:10.1111/j.1365-246X.1961.tb02928.x.

Winklhofer, M., Fabian, K. & Heider, F., 1997. Magnetic blocking temperatures of magnetite calculated with a three-dimensional micromagnetic model, *J. geophys. Res.,* **102**(B10), 22695–22709, doi:10.1029/97jb01730.

Yu, Y., Tauxe, L. & Genevey, A., 2004. Toward an optimal geomagnetic field intensity determination technique, *Geochem. Geophys. Geosyst.,* **5**(2), Q02H07, doi:10.1029/2003gc000630.

Zhao, X., Riisager, P., Riisager, J., Draeger, U., Coe, R.S. & Zheng, Z., 2004. New palaeointensity results from Cretaceous basalt of Inner Mongolia, China, *Phys. Earth planet. Inter.,* **141**(2), 131–140, doi:10.1016/j.pepi.2003.12.003.

Zhu, R., Pan, Y., Shaw, J., Li, D. & Li, Q., 2001. Geomagnetic palaeointensity just prior to the Cretaceous normal superchron, *Phys. Earth planet. Inter.,* **128**(1–4), 207–222, doi:10.1016/s0031-9201(01)00287-4.

Zhu, R., Hoffman, K.A., Pan, Y., Shi, R. & Li, D., 2003. Evidence for weak geomagnetic field intensity prior to the Cretaceous normal superchron, *Phys. Earth planet.Inter.,* **136**(3–4), 187–199, doi:10.1016/s0031-9201(03)00034-7.

Zhu, R., Hoffman, K.A., Nomade, S., Renne, P.R., Shi, R., Pan, Y. & Shi, G., 2004a. Geomagnetic paleointensity and direct age determination of the ISEA (M0r?) chron, *Earth planet. Sci. Lett.,* **217**(3–4), 285–295, doi:10.1016/s0012-821x(03)00613-7.

Zhu, R., Lo, C.-H., Shi, R., Pan, Y., Shi, G. & Shao, J., 2004b. Is there a precursor to the Cretaceous normal superchron? New paleointensity and age determination from Liaoning province, northeastern China, *Phys. Earth planet.Inter.,* **147**(2–3), 117–126, doi:10.1016/j.pepi.2004.01.008.

Zhu, R., Lo, C.-H., Shi, R., Shi, G., Pan, Y. & Shao, J., 2004c. Palaeointensities determined from the middle Cretaceous basalt in Liaoning Province, northeastern China, *Phys. Earth planet.Inter.,* **142**(1–2), 49–59, doi:10.1016/j.pepi.2003.12.013.

Zhu, R., Pan, Y., He, H., Qin, H. & Ren, S., 2008. Palaeomagnetism and 40Ar/39Ar age from a Cretaceous volcanic sequence, Inner Mongolia, China: implications for the field variation during the Cretaceous normal superchron, *Phys. Earth planet.Inter.,* **169**(1–4), 59–75, doi:10.1016/j.pepi.2008.07.025.